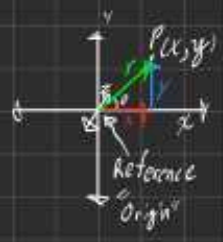
kinematics- the study of motion

The origin of a cartesian System is the Reference.



How do we measure it?
length is a quantity x,y,r
& has direction
Quantity, something you can measure

Quantity: length (L or l), scalar
unit: SI
scalar is a Quantity that scales
magnitude defines scalar
Vector: length
position is a direction defined by length

Scalar · Scalar = Scalar
Scalar · Vector = Vector

Vector × Vector { Scalar / Vector

if it is not vector (has no direction) it is scalar.
- a scalar is simply a value, with or without units or dimension
- π is a dimentionless scalar
- Time is a scalar with dimension

Earths circumference: 40,075,00m = 4E7 (approx)

1st "Where you are" needs a name!
What shall we call where you are?

2nd What things do you need to truly describe where you are?

Coordinate System

Base Quantity

Quantity	Device	Units	SI Units	Vector?
width height length (l)	Rule	in, mm, ft	m	✓
mass (m):	Scale	g	kg	X scalar
Temperature (T):	Thermometer	Fahrenheit	Kelven	X scalar
Time (t):	clock	sec, min, hr	sec	X scalar
Luminous intensity (I)				
Amt of substance (N)				
Current (I)				

Derived quantities come from 2 or more measurments.

Quantity	Dimension	Imp Units	SI Units	Vectors?
Area (A)	length Squared	in^2 ft^2	m^2	✓
Volume (V)	length Cubed	in^3 ft^3	m^3	X scalar
Density	mass/volume		kg/m^3	X scalar
Speed				
Acceleration				
Force				

Conversions, Notations, Estimations, and Pitfalls!

1. How much blood is in the human body (avg)?
   $\approx 5L$

If I say a patient has 1.3gal of Blood.
Are they okay?
$1L = 0.264gal$

We need a conversion factor

Estimate

$1.3gal \cdot \frac{1L}{0.264gal} = 4.92L$

2. Lets say we have 4.92L of olive oil in a Roman Urn.
How full is it?

$1 urna = 13.1L$

$4.92L \cdot \frac{1 urna}{13.1L} = 0.376 urna$

estimate

$\approx 5L \cdot \frac{1 urna}{\approx 15L} =$

3. If we convert L to cubic m, will our # be)
4.92L $1L = 0.001m^3$

$4.92L \cdot \frac{0.001m^3}{1L} = 0.00492m^3$ For Canvas $\rightarrow 4.92 \times 10^{-3}$

$4.92E-3 m^3$

sift 6 → ^

$0.00492m^3 \cdot \frac{1000000cm^3}{1m^3} = 4920cm^3 = 4.92E3 m^3$

Significant Figures (Sig figs)

1. All nonzero digits are significant
   $2.22E4 = 2.22 \times 10^4 = 22200$
   Value
2. Trapped zeros are significant
   $404 = 3sf$, $4004 = 4sf$, $404 = 3sf$, $4004 = 4sf$
   $4004.004 = 7sf$
3. Trailing (on the right) zeros with a decimal are significant
   $4040. = 4sf$, $4040.40 = 6sf$ $4040. \rightarrow 4.040E4$
   $4.04E3 = 3sf$, $4.04E4 = 3sf$
4. Trailing zeros without a decimal are NOT significant
   $4040 = 3sf$, $40400 = 3sf$
5. Leading zeros (on the left) are NOT significant!
   $10 $010 $0.10 $.10

$0.000376004 = 6sf$
$376004E-4$

Scientific notation
4040. 404040
4.040E3 4.04040E3

1A2

6. Precision mathematical Operations
   a. Mult. & Division
      1. Value with the least amount of sig fig

$37 \times 20 = 740 \rightarrow 7E2$

$37 \div 20 = 1.85 \rightarrow 2$

140 2sf $\times 30.2$ 3sf $= 4228 \rightarrow 4.2E3$

$140 \div 30.2 = 4.6357 \rightarrow 4.6$

b. Addition & Subtraction

1. Use the least significant place

32.00006	32.00006	32.00.06	32.00006
+ 4	- 4	+ 4.04	- 4.04
36.00006	28.~~00006~~	36.04006	27.96006
↓	↓	↓	↓
36.	28	36.04	27.96

c. Mixed Mathematical Operations

1. Round inbetween steps

Only when changing rules (+ or ×)

$9.41 \div 3.00$ $\quad 3\overline{)9.41}$

3.13|66

↓

3.14

$12.57 \div 4$ $\quad 4\overline{)12.57}$

3.142|5

↓

3.14

Average length

$$\bar{L} = \frac{1}{N}\sum_{i=1}^{N=4} L_i = \frac{1}{4}[L_1 + L_2 + L_3 + L_4] = \bar{x}$$

Standard deviation (std. dev.)

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\bar{x}-x_i)^2}$$

sigma

$$= \sqrt{\frac{1}{N-1}\left[(\bar{x}-L_1)^2+(\bar{x}-L_2)^2+(\bar{x}-L_3)^2+(\bar{x}-L_4)^2\right]}$$

SEM - Standard Error of the Mean

$$\alpha = \frac{\sigma}{\sqrt{N}}$$

σ: std. dev.

N: # of measurments

When aiming for high precision in calculations round as you go! Each time you switch between sig. Rules. e.g. (+ −) or (× ÷)

Example:

Find the avg. of student measurments.

#1 = 4.5cm
#2 = 4.34cm
#3 = 4.67cm
#4 = 4.3cm

Step1. Add:

$$\begin{array}{r} 4.5 \\ 4.34 \\ 4.67 \\ +\ 4.3 \\ \hline 17.81 \end{array}$$

sig. ↓ 17.8 cm

Step2. Divide

Avg = Sum ÷ # of measurments

= 17.8cm ÷ 4

= 4.45cm

3 sig? ↓

4.45cm

Why did I not round to 1 sig?

Ans

Remember! Counting #s do not factor into sig fig determinations because there no ambiguity in the number.

There are exactly 4 Students who made exactly 4 measurments.

Remember, do not round pi when using it in calcs.

Isaiah Jimenez

Fractional Uncertainty

$$\frac{\text{uncertainty}}{\text{mean}} = \frac{a}{\bar{x}}$$

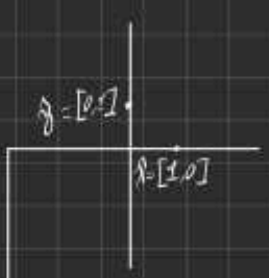
Notation: Parentheses (Coordinate) vs Bracket (Vector)

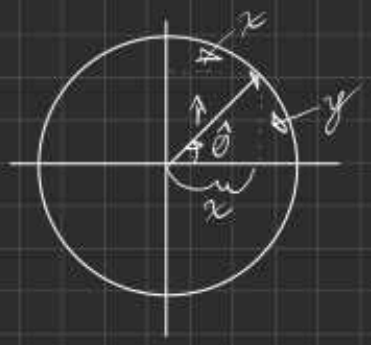
Vector, Tail, Tip
[0,1],(0,0),(0,1)

Vector, tail, tip
[2,2],(0,0),(2,2)

Vector, tail, tip
[1,0],(0,0),(1,0)

Vector, tail, tip
[2,-2],(0,-2),(2,0)





$x = r\cdot\cos(\theta)$
$y = r\cdot\sin(\theta)$

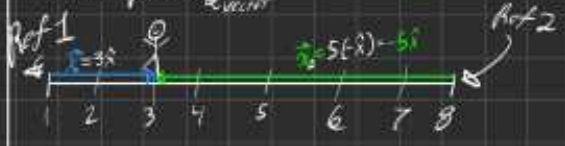
$\sin=\frac{o}{H}$ $\cos=\frac{a}{H}$ $\tan=\frac{o}{a}$

$r=\sqrt{x^2+y^2}=hyp.$

All vector spaces are defined by unit vectors!
- Parallel to Cartesian axes
- Magnitude = 1

Unit vectors define direction of something/anything!

Like position vector



Unit vectors can be negative

Transformations Vs. Translation

Addition, Subtraction & Commutativity

1. Addition can only occur if vectors have the same physical & spatial dimension.

$\vec{x}_1+\vec{x}_2=\vec{x}_3$ $\quad$ $\vec{x}_1+\vec{x}_2=\vec{x}_3$

$\vec{x}+\vec{u}=?$ $\quad$ $\vec{y}_1+\vec{y}_2=\vec{y}_3$

$\vec{x}+\vec{y}=?$

2. Zero & itself

$\vec{x}+0=\vec{x}$ $\quad$ $[2,2]+0=[2,2]$

$\vec{x}_1+\vec{x}_1=2\vec{x}_1$ $\quad$ $(2\hat{x}+2\hat{y})+0=2\hat{x}+2\hat{y}$

$[2,2]+[2,2]=[4,4]$

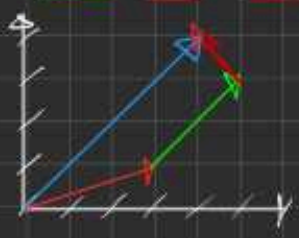
$(2\hat{x}+2\hat{y})+(2\hat{x}+2\hat{y})=4\hat{x}+4\hat{y}$

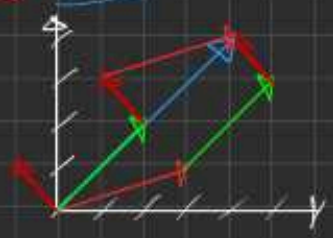
Resultant

3. Can change both magnitude & direction

$[2,2]+[3,1]=[5,3]$

$[2,2]+[3,1]+[-1,1]=[4,4]$





Commutativity

$\vec{V}_1+\vec{V}_2=\vec{V}_2+\vec{V}_1$ Commutes!

Order doesn't Matter

$\vec{V}_1-\vec{V}_2\neq\vec{V}_2-\vec{V}_1$

Does not commute order matters

$2\hat{x}+2\hat{y}$ + $3\hat{x}+1\hat{y}=5\hat{x}+3\hat{y}$

$[2,2]+[3,1]=[5,3]$

$R_1=\vec{V}_1+\vec{V}_2=V_2+\vec{V}_1$

$[2,2]-[3,1]=[-1,1]$

$[3,1]-[2,2]=[1,-1]$

$R_1=\sqrt{5^2+3^2}=\sqrt{34}$

Commutativity

$\vec{V}_1+\vec{V}_2=\vec{V}_2+\vec{V}_1$ (Commutes)

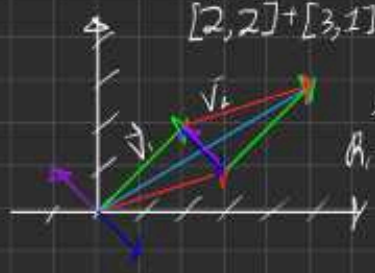
Order doesn't Matter

$\vec{V}_1-\vec{V}_2 \neq \vec{V}_2-\vec{V}_1$

Does not commute order matters

$2\hat{i}+2\hat{j} + 3\hat{i}+1\hat{j} = 5\hat{i}+3\hat{j}$

$[2,2]+[3,1]=[5,3]$



$R_1=\vec{V}_1+\vec{V}_2=V_2+\vec{V}_1$

$[2,2]-[3,1]=[-1,1]$

$[3,1]-[2,2]=[1,-1]$

Polar

$\vec{V}_1=[2,30^\circ]$

$[\sqrt{3},1]$

$[1.7,1]$

$\vec{V}_2=[3,60^\circ]$

$\left[\frac{3\sqrt{3}}{2},\frac{3}{2}\right]$

$[2.6,1.5]$

$x_1=r\cdot\cos(\theta)$

$=2\cdot\cos(30)$

$=\frac{2\sqrt{3}}{2}$

$x=\sqrt{3}$

$y_2=r\cdot\sin(\theta)$

$3\cdot\sin(60^\circ)$

$=3\ \frac{\sqrt{3}}{2}$

$y=\frac{3\sqrt{3}}{2}$

$y=r\cdot\sin(30)$

$y_1=2\cdot\frac{1}{2}$

$y=1$

$x_2=r\cos(\theta)$

$x=3\cdot\cos(60)$

$x=3\cdot\frac{1}{2}$

$\vec{V}_1=(2\cos 30^\circ)\hat{x}+(2\sin 30^\circ)\hat{y}$

$\vec{V}_2=(3\cos 60^\circ)\hat{x}+(3\sin 60)\hat{y}$

$\vec{V}_1+\vec{V}_2=\sqrt{3}\,\hat{x}+1\hat{y}+\frac{3}{2}\hat{x}+\frac{3\sqrt{3}}{2}\hat{y}$

$=(\sqrt{3}+\frac{3}{2})\hat{x}+(1+\frac{3\sqrt{3}}{2})\hat{y}$

$=3.252\hat{x}+3.598\ \hat{y}$

$R=\sqrt{(3.252)^2+(3.598)^2}$

$R\approx 4.85$

Polar$(4.85, 48^\circ)$

$\theta=\tan^{-1}\left(\frac{3.598}{3.252}\right)$

$\theta\approx 48^\circ$

Multiplication

1 By zero: $\vec{V}*0=0$ $(2\hat{i}+2\hat{j})*0=0$

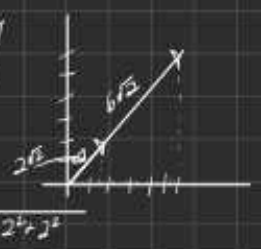
$[2,2]*0=0$

$\vec{V}*1=\vec{V}$

2. By positive scalars

$[2,2]*1=[2,2]$

$[2,2]*3=[6,6]$



$R=\sqrt{6^2+6^2}$

$=\sqrt{36\cdot 2}$

$=6\sqrt{2}$

3. By negative scalars

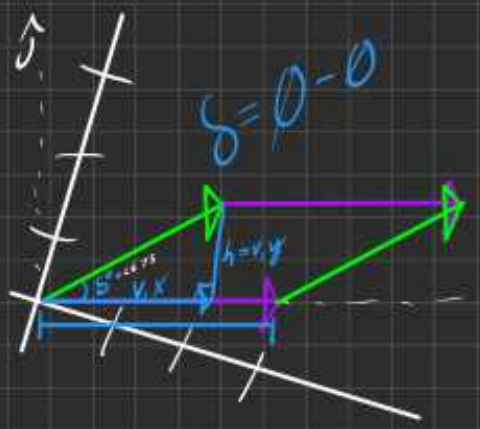
$[2,2] * -3 = [-6,-6]$

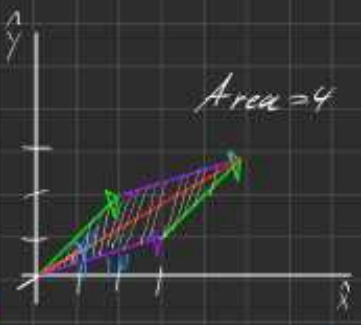
4. By any other vector or array of vectors changes magnitude, direction, and/or dimension!

$\vec{I}_1 * \vec{I}_2$ (1D 1D) or $\vec{I} * \vec{W}$ (1D 1D) → Area 2D $= \vec{A}$

?

$[2,2] * [3,1] = [3,4,2]$





$\theta - \theta$

V_2 angle to x

V_1 angle to x

Vectors Row Vs Column

$[2,2]$ (x y) $\quad \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ (x y)

$[3,1] \quad \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$[2,2] * \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 6+2 = 8$? is this area?

A = b h

$V_1 = [2,2] \qquad \theta = \tan^{-1}\frac{2}{2} = 45°$

$V_2 = [3,1] \qquad \theta = \tan^{-1}\left(\frac{1}{3}\right) = 18.43°$

$\delta \; 26.75°$

$\sin\delta = \frac{h}{V_1} \qquad V_1 = \sqrt{2^2+2^2} = 2\sqrt{2}$

$h = V_1 \sin\delta = 2\sqrt{2}\,\sin 26.57° = 1.265$

$b = |V_2| = \sqrt{3^2+1^2} = \sqrt{10}$

$$\boxed{A = b \cdot h = \sqrt{10} \cdot 1.265 = 4}$$

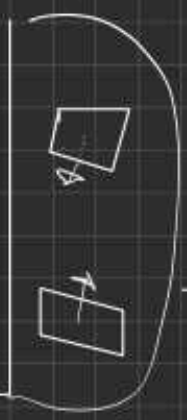
Area (Array)

$[2,2] \quad [3,1]$

$\begin{bmatrix} 2 & 2 \\ 3 & 1 \end{bmatrix} \Rightarrow \begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix} = 2-6 = -4$

Matrix

$\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \Rightarrow \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} = 6-2 = 4$



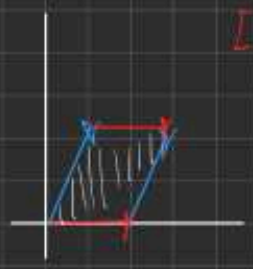
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ = Identity Matrix

$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0+0 = 0$

$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1+0 = 1$ ← Area

$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0+1 = -1$

Shear / skew



$[1\ 0]$ $[0\ 3\ 1]$

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} = 1\hat{z}$$

If vectors are parallel NO Area!

$[2,2,0]$ $[3,1,0]$

$$\begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{bmatrix}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = \hat{x}(0-0) - \text{ or } \hat{x}\begin{vmatrix} 2 & 0 \\ 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = -\hat{y}(0-0) + \text{ or } \hat{y}\begin{vmatrix} 2 & 0 \\ 3 & 0 \end{vmatrix}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = +\hat{z}(2-6) \text{ or } \hat{z}\begin{vmatrix} 2 & 2 \\ 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = \hat{x}(0-0) - \hat{y}(0-0) + \hat{z}(2-6) = -4\hat{z}$$

Find the area of a Δ w/ vertices: $(-1,-2), (2,-1), (1,3)$

Tip Tail

$\vec{V}_1 = (1-(-1))\hat{x} + (3-(-2))\hat{y}$

$V_1 = 2\hat{x} + 5\hat{y} = [2,5]$

$\vec{V}_2 = (2-(-1))\hat{x} + (-1-(-2))\hat{y}$

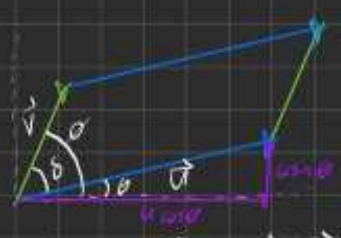
$= 3\hat{x} + 1\hat{y} = [3,1]$

$[2\ 5\ 0]$ $[3\ 1\ 0]$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 5 & 0 \\ 3 & 1 & 0 \end{vmatrix} = 2-15\hat{z} = -13\hat{z} \text{ Area of area.}$$

$\Delta\text{area} = \frac{1}{2} \cdot A = \frac{1}{2}(-13\hat{z}) = -\frac{13}{2}\hat{z}$

Geometric Dot vs Cross



What if? $\vec{u} * \vec{v}$?

$\vec{u} = u\cos\theta\hat{i} + u\sin\theta\hat{j}$

$\vec{v} = v\cos\phi\hat{i} + v\sin\phi\hat{j}$

$\vec{u} * \vec{v} = (u\cos\theta\hat{i} + u\sin\theta\hat{j})(v\cos\phi\hat{i} + v\sin\phi\hat{j})$

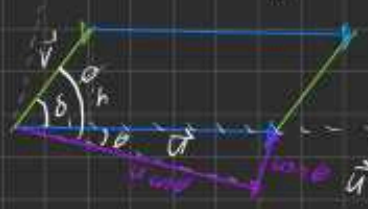
$= uv\cos\theta\cos\phi(\hat{i}\cdot\hat{i}) + uv\sin\theta\cos\phi(\hat{j}\cdot\hat{j})$

$+ uv\cos\theta\sin\phi(\hat{i}*\hat{j}) + uv\sin\theta\cos\phi(\hat{j}*\hat{i})$

ugh, lets simplify

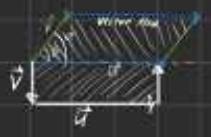
$\sin 0 = 0$ ∴

$\cos 0 = 1$



$\vec{u} * \vec{v} = \underbrace{uv \cdot 1 \cdot \sin\phi(\hat{i}*\hat{j})}_{\text{Area}} + \underbrace{uv \cdot 1 \cdot \cos\theta(\hat{i}\cdot\hat{i})}$

$uv\cos\theta(\vec{u}\cdot\vec{v})$



Scalar Region

Dot Product: $uv\cos\delta$ while $\delta \le 90°$

$$[u_x \ u_y \ u_z]\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

gives scalar

$$\boxed{\vec{A} = \vec{u} \times \vec{v} = uv\sin\delta}$$

Area

Definition of Cross Product (curl)

Polar Vector

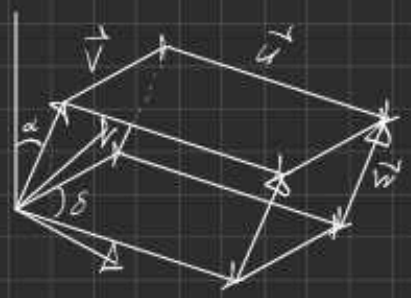
$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$ ← Cartesian

If you have...

or preference

gives vector

Volume



$V = \vec{L} \times \vec{L} \cdot \vec{L}$

$V = \vec{A} \cdot \vec{L}$

$V = \overbrace{uv\sin\delta\,\hat{z}} \cdot \vec{w}\,\hat{z}$

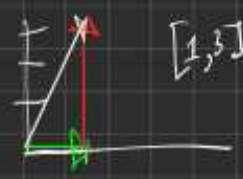
$= uvw\sin\delta\cos\alpha$

$V = \vec{A} \cdot \vec{L} = \vec{u} \times \vec{v} \cdot \vec{w}$

$$= \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{bmatrix}\begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{vmatrix} w_x & w_y & w_z \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}$$

$$\text{or} = [A_x \ A_y \ A_z]\begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$$

Unit vectors have magnitude of 1.



$[1,3]$

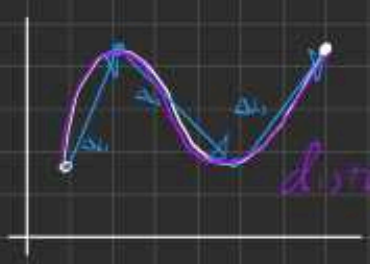
u_x = vector $[1,0]$

Ex

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{vmatrix} = (0-0) - (0-0) + (2-6) = -4 \text{ units}^3$$

1.B.2 - Last tidbits

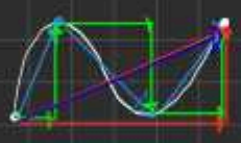
Curvilinear Lengths!



Estimate $L = \Delta L_1 + \Delta L_2 + \Delta L_3$

distance

scalar



$L \approx \vec{\Delta L_1} + \vec{\Delta L_2} + \vec{\Delta L_3}$

$L \approx \sum L_x\hat{x} + \sum L_y\hat{y}$

$\approx \sqrt{\sum L_x\hat{x} \cdot \sum L_y\hat{y}}\ \hat{r}$

displacement

Vector

$\Delta L \to dl$

Continuous addition of tiny pieces of length → Line integral.

cartesian → $dl = dx\,\hat{x} + dy\,\hat{y}$

polar

$$dl = \sqrt{dx^2 + dy^2} = \sqrt{\left(\frac{dx^2}{dx^2} + \frac{dy^2}{dx^2}\right)dx^2} = \sqrt{1 + \frac{dy^2}{dx^2}}\,dx$$

$$\int dl = \int_{L_1}^{L_2} \sqrt{1 + \frac{dy^2}{dx^2}}\,dx$$

$$L = \int_{L_1}^{L_2} \sqrt{1 + \frac{dy^2}{dx^2}}\,dx$$

Circumference

$x = r\cos\theta \qquad y = r\sin\theta$

$dx = dr\cos\theta - r\sin\theta\,d\theta \qquad dy = dr\sin\theta + r\cos\theta\,d\theta$

$$ds \approx dl = \sqrt{dx^2 + dy^2}$$

$$= \sqrt{(dr\cos\theta - r\sin\theta\,d\theta)^2 + (dr\sin\theta + r\cos\theta\,d\theta)^2}$$

$$= \sqrt{dr^2\cos^2\theta + r^2\sin^2\theta\,d\theta^2 - \cancel{2dr\sin\theta\cos\theta\,d\theta} + dr^2\sin^2\theta + r^2\cos^2\theta\,d\theta^2 + \cancel{2dr\sin\theta\cos\theta\,d\theta}}$$

$$\sqrt{dr^2 + r^2 d\theta^2(\sin^2\theta + \cos^2\theta)}$$

$dl = \sqrt{dr^2 + r^2 d\theta^2}$

circle radius does not change

$ds = r\,d\theta$

$\int ds = r\int_0^{2\pi} d\theta$

$s = r\theta\Big|_0^{2\pi}$

$\boxed{s = 2\pi r}$ circumference

Area using infinitessimals

$dA_{ring} = C \cdot width$

$dA = 2\pi r\,dr$

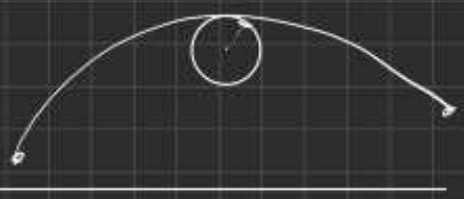
$\int dA = \int_0^R 2\pi r\,dr$

$A = 2\pi\int_0^R r\,dr$

$A = 2\pi\left[\frac{r^2}{2}\right]_0^R$

$\boxed{A = \pi R^2}$

Wount work for a projectile



Jacobian

What: Jacobian matrix basically describes the amount of stretching, rotating, or transforming. In other words, for every point of a function, what is happening locally?

Why? Sets up area & volume elements to help with finding moment on inertia, electric/ magnetic flux &macro fields (physics II) along with Gauss & Ampere's Laws

Consider Cartesian to Polar $(x, y) \to (r, \theta)$

Take correlated but independent measurements x and y

$x = r\cos\theta \qquad y = r\sin\theta$

the determent gives us the scaling factor.

$$\begin{bmatrix} \frac{dx}{dr} & \frac{dx}{d\theta} \\ \frac{dy}{dr} & \frac{dy}{d\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} = r\cos^2\theta + r\sin^2\theta = r(\cos^2\theta + \sin^2\theta) = r$$

$dA = dx\,dy = \frac{Det\ J}{F} \cdot dr\,d\theta$

Try to use these

$$\int dA = \int_0^{2\pi}\int_0^R r\,dr\,d\theta$$

$$A = \int_0^{2\pi} \frac{R^2}{2}\,d\theta$$

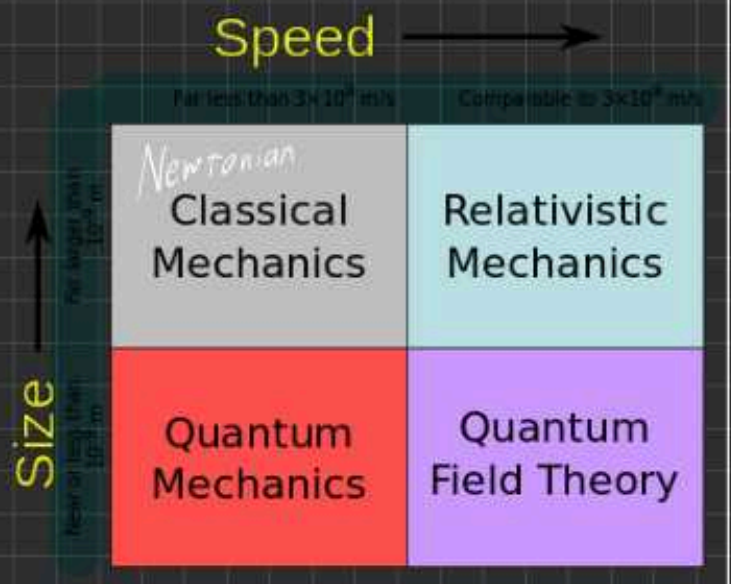
$$A = \frac{R^2}{2}\int_0^{2\pi} d\theta$$

$$A = \frac{R^2}{2}\cdot 2\pi$$

$$A = \pi R^2$$

Pre Lab Questions Section

Part II: Classical Mechanics



Definitions:

1. Definition of Matter Mass
   "The quality of matter is the measure of the same, arising from its density and bulk conjointly."
   $D = \frac{m}{V}$ Mass = ρ · Volume
2. Concerns motion
   Where you are vs, where you are going

   Velocity - cares about
   Speed - does not care about
   Acceleration - change in speed
   2nd derivative of position
   gives concavity
   Jerk - change in acceleration
   2nd derivative of velocity
   concavity of velocity
   snap - change in jerk
   Distance vs displacement

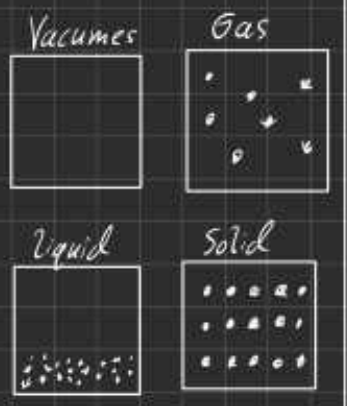
displacement: is a vector

Avg

$a = \frac{\Delta V}{\Delta t}$ $a = \frac{dv}{dt}$

$V = \frac{\Delta x}{\Delta t}$ $V = \frac{dx}{dt}$

Mediums!



Vacuums: $\rho \approx 0$

Gasses: ρ is low
Compressibility high

Liquids: Gas < ρ < solids
compressibility lower than gas

Solids:

Nerve impulses travel a a speed of 100 m/s. If you accidentally stub your toe in the dark, estimate how much time it takes for the signal to reach your brain.

This question requires you to make an assumption. What is it?

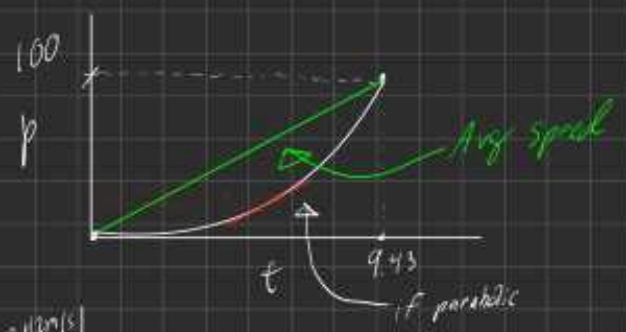
$V = \frac{\Delta x}{\Delta t} = 200 \frac{m}{s}$

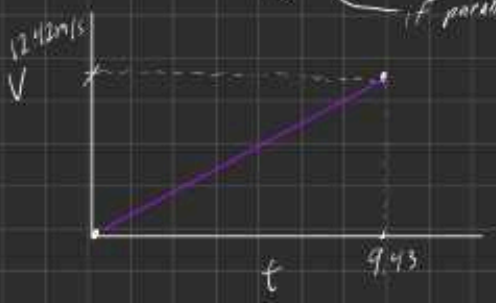
Usain Bolt hold the record of 9.58 s for the 100 m sprint, with a record time of 0.148 s. His top speed is 27.79 mph.
1 m/s = 2.237 mi/hr

$$Avg = \frac{100m}{(9.58s - 0.148s)} \cdot \frac{2.237mi}{1hr} = 23.7mi/hr$$

$$Max = 27.79\,mph \cdot \frac{m/s}{2.237mph} = 12.42m/s$$

$$Avg = 10.6\,m/s$$

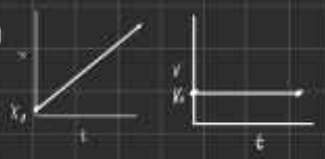




$x = \frac{1}{2}at^2 + V_0t + X_0$
$V = at + V_0$

a = constant

Uniform Motion
$a = 0$
$a \approx 0$
a = being ignored



$a = 0$ V = constant = V_0 $x = V_0t + X_0$
height or initial position

Uniformly accelerated motion
$a = a_0$ $V = a_0t + V_0$ $X = \frac{1}{2}at^2 + V_0t + X_0$

$\int a\,dt = \int a_0\,dt$

$V = a_0t + C$
m/s s

$X - X_0$
$\Delta Y = \frac{1}{2}at^2 + V_0t$

The avg person passes out at an acceleration of 70.0 m/s^2. Suppose a car is designed to accelerate at this rate (would need rocket boosters). How much time is required for the car to accelerate from rest to 60.0 mph?

$$70.0\,\frac{m}{s^2} = \frac{2.237mph}{1m/s} = 156.1mph/s$$

$V = at + V_0 \Rightarrow \Delta V = at$

$a = \frac{\Delta V}{t}$ where $t_0 = 0$

$$156.1mph/s = \frac{60.0mph}{t}$$

$$t = 0.384s$$

Rockin Roller Coaster

This coaster at WDW's Hollywood studios sends rides from rest to 60. Mph in 2.8 s. Calculate the length of track needed to achieve this.

$$60.mph \cdot \frac{1\,m/s}{2.237mph}$$

$$a = \frac{\Delta V}{\Delta t} = \frac{26.8m/s}{2.8}$$

$$\boxed{a = 9.57m/s^2}$$

$$\Delta x = \frac{1}{2}(9.57\,m/s^2)(2.8s)^2 + 0$$

$$\Delta x = 38m \cdot \frac{3ft}{1m} \approx \boxed{120ft}$$

$\vec{V} \parallel \Delta\vec{x}$
Velocity will alway be parallel to displacement

a { speeding up $V \parallel a$
slowing down $V \parallel a$ anti parallel

Theory of Impetus

Is an objects motion persistent?

friction dissipates motion

Definition II: The quantity of motion is the measure of the same, arising from the velocity and quantity of matter conjointly.

$m\vec{V} = \vec{p}$

momentum

Center of mass

Elaboration: The motion of the whole is the sum of the motions of all the parts; and therefore in a body is quantity, with equal velocity, the motion is double; with twice the velocity, it is quadruple.

$4p = 2m\,2V$

quantity of matter is mass
quantity of is momentum

Definition III: Vis Instia - Innate "force" of matter
The vis insta, or innate force of matter, is a power of resisting, by which every body, as much as in it lies, endeavors to preserve in its present state, whether it be at rest, or of moving uniformly forward in a right line.

Inertia is proportional to the bodies mass

Inertia - at rest
in motion

Definition IV: Changing Momentum - Impressed Forces

An impressed force is an action …

the action no longer remains in the body, when the action is over. and maintains its new state. By Inertia

force
Origins: percussion, pressure, centripetal

Percussion: forces that are impulsive, often with short timescales of application

Pressure: forces distributed over an area

Centripetal: forces that develop or maintain angular motion.

$$\vec{F} \propto \Delta\vec{p}$$

Law 1: Law of Inertia

Every body preserves in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed thereon.

Net Force - the vector sum of all external forces exerted on an object.

Equilibrium - if sum of forces = 0

Static: $\vec{V} = 0$ $\vec{a} = 0$ $\sum \vec{F} = 0$ Rest

Dynamic: $\vec{V}$ = Constant $\vec{a} = 0$ $\sum \vec{F} = 0$ Uniform Motion

$\sum \vec{F} \neq 0 \rightarrow$ Out of equilibrium

1. 1D Forces
2. Add Fs up

$\sum F = \vec{F}_1 + \vec{F}_2 + \ldots$

Law II & Corollary I

The alteration of motion is ever proportional to the motive force impressed: and is made in the direction of the right line in which that force is impressed.

$$\vec{F} \propto \Delta\vec{p} = \Delta(m\vec{V})$$

$$F = m\frac{\Delta\vec{V}}{\Delta t} = m\vec{a}$$

Newton's 3rd Law

"To every action there is always opposed and equal reaction: or the mutual actions of two bodies upon each other are always equal, and directed to contrary parts."

Forces:

Fundamentals — Nuclear Forces { strong, weak

Physics 1 → Gravity (Field)

Physics 2 → Electromagnetic (Field)

' Contact vs Non-Contact

That there is a power of gravity tending to all bodies, proportional to the several quantities of matter which they contain.

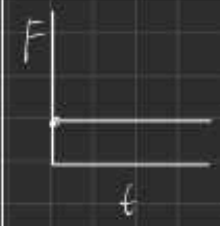
That all bodies gravitate towards every planet; and that the weights of bodies towards any the same planet, at equal distances from the center of the planet, are proportional to the quantities of matter which they generally contain.

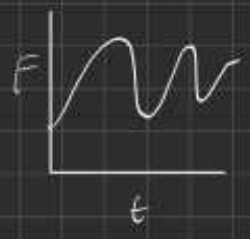
That the force by which the moon is retained in its orbit tends to the earth; and its reciprocally as the square of the distance of its place from the earth's center.

$F \propto \frac{1}{r^2}$

That force of gravity, considered downward form the surface of the planets, decreasing nearly In the proportion of the distances from the counters of the matter of the planet were of a uniform density, ...

If any force generates a motion, a double force will generate double the motion, and so on.





Corollary I: A body by two forces conjoined will describe the diagonal of a parallelogram, in the same time that is would describe the sides, by those forces apart.

Superposition definition:
For all systems that can be modeled linearly, the net response (ie vector sum), at a given place and time, caused by two or more stimuli (forces) is the sum of the responses (vectors) which would have been caused by each stimulus (force) individually.

Since force is always directly linearly, it obeys superposition, and the sum of all forces acting upon matter can be added to develop a magnitude of net force, if this magnitude is zero there is either static (V= 0) or dynamic equilibrium (v = constant)

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 \ldots = ma$$

$$\sum \vec{F} = \vec{ma}$$

$$\vec{V}_1 + \vec{V}_2 =$$

IPCC-reports

Gravitational Force = Weight ≠ mass

$F_g = G \frac{M_1 M_2}{r^2}$

Gravitational constant

$F_g = 6.67\times10^{-11} \frac{m^3}{kg\,s^2} \cdot \frac{M_{me} M_E}{r^2}$

$F_g = 6.67\times10^{-11} \frac{m^3}{kg\,s^2} \cdot \frac{M_{me}(5.97\times10^{24}\,kg)}{(6.378\times10^6\,m)^2}$

$\frac{kg\,m}{s^2} = N = F_g = 9.81 \frac{m}{s^2} M_{me}(kg)$

$F_g = Ma$

Vector

$r = 6.378\times10^6\,m + (408 + 1000) = 6.786\times10^6$

$F_g = 6.67\times10^{-11} \frac{m^3}{kg\,s^2} \cdot \frac{M_{me}(5.97\times10^{24}\,kg)}{(6.786\times10^6\,m)^2}$

$F_g = 8.6 \frac{m}{s^2}$

Newton's 3rd law

"To every action there is always opposed and equal reaction: or the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

Force earth on me = mass me * acceleration me = - force me on earth

Force is mutual!

$F_{E\,me} = M_{me} a_{me} = -F_{me\,E}$

$a_E = -\frac{M_{me}}{M_e} g \approx 0$

Conversion Factors

F_g = weight ≠ Mass

W

lb

$1\,lb \cdot \frac{4.448\,N}{1\,lb} = 6227\,N$

$4.448\,N = 1\,lb$

$N = kg \frac{m}{s^2}$

$\frac{6227\,kg\,m/s^2}{9.81\,m/s^2} = 63.48\,kg$

$0.453\,kg = 1\,lb$

or

$140\,lb \cdot \frac{0.453\,kg}{1\,lb} = 63.42\,kg$

G-Force! Little g vs Big G

an acceleration

$g = 9.81 \frac{m}{s^2} = 1G$

$70 \frac{m}{s^2} \cdot \frac{1G}{9.81 \frac{m}{s^2}} = 7.14G$

The Expanse

TV/Movie

did good

John Carter

$g = 9.81 \frac{m}{s^2} = 1G$

Roller Coasters: Avg: 5G

peak: 7G

Pass Out: 7G Avg

Death: depends on direction Avg 15G

$P = \frac{F}{A}$

The cohesion of matter

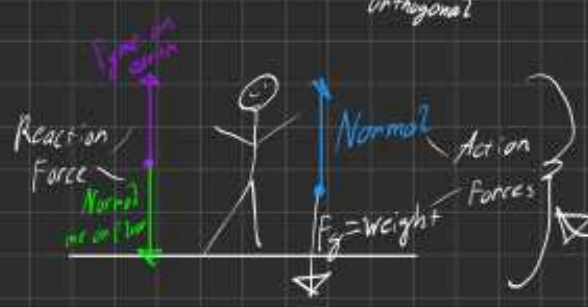
Depends of the type of matter you are talking about.

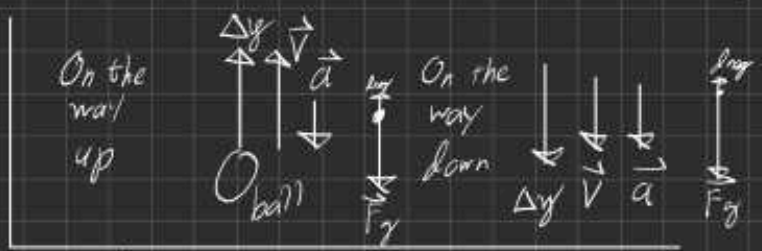
Vaccum——>Gas——>liquid——>Solid

Normal

⊥

orthogonal





free body diagram

↓

a/a

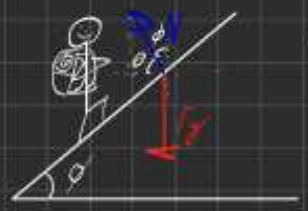
↓

Force diagram / should be seperate from kinemat diagram

Stationary Reference Frames

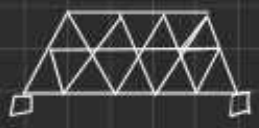
Action: $\sum F = ma = 0$

$N - F_g = 0$



$\sum F = 0 = (-N\sin\theta)\hat{x} + (N\cos\theta - F_g)\hat{y}$

$\cos\theta$ or $\sin\theta$



Reference frames

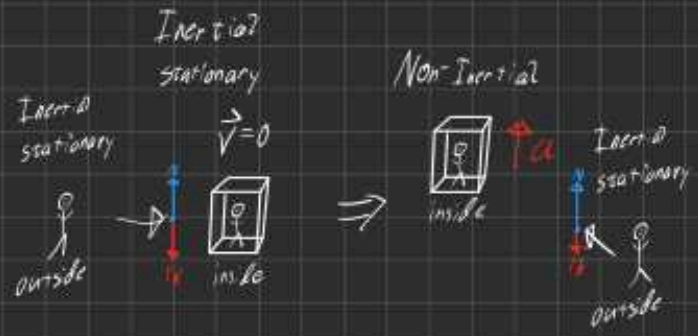
Stationary:
- Observations made from observers $\vec{v}=0$; $a=0$

Absolute:

Moving → Inertial: $\vec{V}$ = constant; $a=0$

Moving → Non-Inertial $a \neq 0$
- Gives rise to "fictitious" forces or "Inertial"

Not real forces

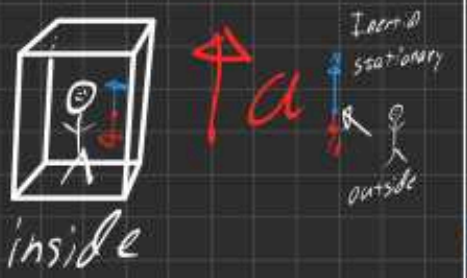


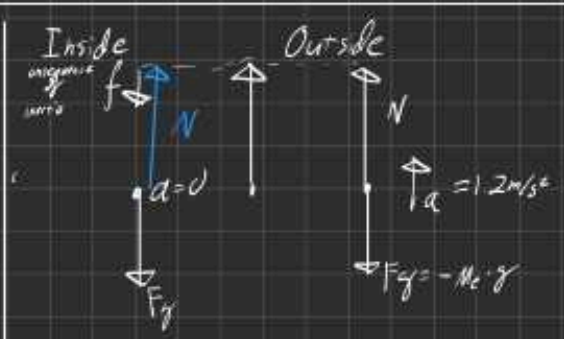
1st Law Inertia

2nd Law $\vec{F} = m\vec{a}$

3rd Law

Non-Inertial





$\sum F = N - F_g - f$

$f = 698N - F_g$

$= 698N - (M_c \cdot g)$

$f = 698N - 622N$

$f = 76.1N \cdot \frac{1lb}{4.48N}$

$f = 17.1lb$

$\sum F_{cart} = N - F_g = M_c a$

$N = M_c a + M_c g$

$= M_c(a + g)$

$= 63.2(1.2 + 9.81)$

$N = 698N$

160ft at 45mph

start 921ft → 1,081ft

calculate accel.

$a = \frac{\Delta v}{\Delta t}$ $\quad v = \frac{\Delta x}{\Delta t}$

$a = \frac{20.1 m/s}{2.43 s} = 8.27 m/s^2$

$t = 48.8m \cdot \frac{1s}{20.1m} = 2.43s$

$\frac{8.27m}{s^2} \cdot \frac{s^2}{9.81m} = 0.84G$

$45mi \cdot \frac{5280ft}{1mi} = 237600 ft/hr \cdot \frac{1hr}{60min} \cdot \frac{1min}{60sec} = 66 ft/s \cdot \frac{0.3048m}{1ft} = 20m/s$

Uniform acceleration

Use → $x = \frac{1}{2}at^2 + v_0 t + x_0$

$160ft \cdot \frac{0.3048m}{1ft} = 48.8m$

$4G \cdot 9.81\,\frac{m}{s^2} = 39.24\,m/s^2$

$a = \frac{\Delta V}{\Delta t}$

$39.24\,m/s^2 \cdot 2.43\,s = \Delta V$

$95.4\,m/s = \Delta V$

Δx

$V_0 = 0$

$V_f = 20.1168\,m/s$

$V^2 = \cancel{V_0^2} + 2a\Delta x$

$4 \cdot 9.81 \cdot \Delta x$

$\Delta x = \frac{V^2}{2a} = \frac{(20.1168\,m/s)^2}{2 \cdot 4 \cdot 9.81\,m/s^2} = 5.16\,m$

$V_f = at + \cancel{V_0}$

$t = V_f / a = \frac{20.1168\,m/s}{4 \cdot 9.81\,m/s^2}$

$t = 0.512\,s$

Safety

$F_g = \text{weight} = mg = 1kg \cdot 9.81\,m/s^2$

$= 9.81\,N$

$= 2.2\,lb$

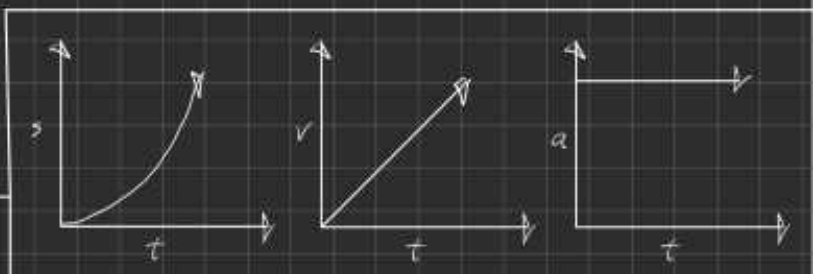
$\Sigma F = N - F_g = ma$

$N = ma + mg$ $\quad a = 4g \cdot 9.81$

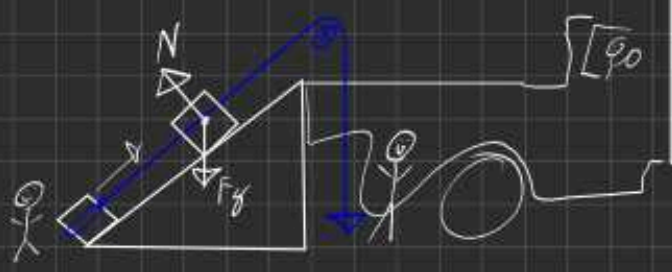
$N = M(a+g)$ $\quad a = 39.24\,m/s^2$

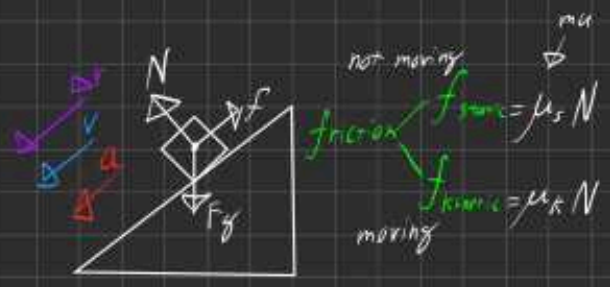
$N = 1kg(39.24 + 9.81)$

$N = 49.05\,N \quad \frac{1\,lb}{4.448\,N} = 11.03\,lb$



$x = \frac{1}{2}a(t)^2 + V_0(t) + x_0$ $\quad V_f = a(t) + V_0$ $\quad a = a_0$





$f_{static} = \mu_s N$

$f_{kinetic} = \mu_k N$

$\mu_k < \mu_s$

The direction of friction is always opposed to expected motion.

$\Sigma F = ma$

$= (f - F_g \sin\theta)\hat{i} + (N - F_g\cos\theta)\hat{j} = M(a\hat{i} + a\hat{j})$

$ma\hat{j} = 0$

No accel in the j direction

$(f - F_g \sin\theta)\hat{i} = ma\hat{i}$ & $N - F_g\cos\theta = 0$

$\mu N - Mg\sin\theta = ma\hat{i}$ & $N = mg\cos\theta$

$\mu mg\cos\theta - Mg\sin\theta = ma\hat{i}$

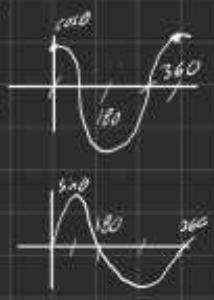
$\mu mg\cos\theta - mg\sin\theta = ma$

$$\boxed{a = g(\mu\cos\theta + \sin\theta)}$$

@ 90° is free fall

$a = g(\cancel{\mu\cos 90} + \sin 90)$

$a = +g$



$a = g(\mu\cos\theta - \sin\theta)$

@ $\theta = 0°$

$a = g(-\mu\cos 0° - \sin 0°)$
$a = -g\mu$

$a \leftarrow \boxed{} \rightarrow f = \mu N$

m_2: $\sum F + T - m_2 g = -m_2 a$

$\sum F = ma$

m_1: $(T - f - F_g\sin\theta)\hat{i} + (N - F_g\cos\theta)\hat{j} = m_1(a\hat{i} + \cancel{a\hat{j}})$

m_1: $T - \mu N - m_1 g\sin\theta = m_1 a$ } $N = m_1 g\cos\theta$

Simplification: No M string
No f pulley
No stretch of string
$|a_1| = |a_2|$ $T_1 = T_2$

m_1: $T - \mu m_1 g\cos\theta - m_1 g\sin\theta = m_1 a$

M_2: $T - m_2 g = -m_2 a$
$T = m_2 g - m_2 a$

M_2: $m_2 g - m_2 a - \mu m_1 g\cos\theta - m_1 g\sin\theta = ma$

$m_2 g - \mu m_1 g\cos\theta - m_1 g\sin\theta = m_1 a + m_2 a$
$g(m_2 - \mu m_1\cos\theta - m_1\sin\theta) = a(m_1 + m_2)$

$a = \dfrac{g(m_2 - \mu m_1\cos\theta - m_1\sin\theta)}{m_1 + m_2}$

$a = 0 = \dfrac{g(m_2 - \mu m_1\cos\theta - m_1\sin\theta)}{m_1 + m_2}$

$a = 0 = \dfrac{g(m_2 - \mu m_1\cos\theta - m_1\sin\theta)}{m_1 + m_2}$

$a = 1.39\,m/s^2$

$a = \dfrac{g(1 - 0.2\cdot 2\cos 60 - 2\sin 60)}{2 - 1.9}$

$a = -X$

0 means equilibrium

If this system = 0 then accel = 0

$0 = m_2 - \mu m_1\cos\theta - m_1\sin\theta$

$m_2 = m_1(\mu_s\cos\theta + \sin\theta)$

let $\theta = 30°$, $\mu = 0.5$

$m_2 = m_1(0.5\cdot\frac{\sqrt{3}}{2} + \frac{1}{2})$

$m_2 = m_1(\frac{\sqrt{3}}{4} + \frac{1}{2})$

$m_2 = m_1(0.93)$

$m_2 = 93\%$ of m_1 for equilibrium

To break static friction $m_2 > 93\%$ of m_1

vector

Drag Force

Air/liquid → hydrodynamic
↳ Aerodynamic

$\vec{D} \propto \vec{v}^2$

$\vec{D} \propto \vec{A}$ Surface area in direction of travel

$\vec{D} \propto \rho$ (density)

$\rho_{air} < \rho_{fresh\ water} < \rho_{salt\ water}$

$\rho_{straw} < \rho_{tapo}$

$\vec{D} \propto$ Drag Coefficient C

$\rho_{warm\ blood} < \rho_{cold\ fry}$

$\boxed{\vec{D} = \frac{1}{2}\rho A C v^2}$

relative

This is only an estimate.

NEOs
near earth objects

Lab blocks: m_1 & m_2
$a = 1.39\,m/s^2$

$\vec{D} = \frac{1}{2}\rho A C \boxed{v^2}$

$\vec{D} = \frac{1}{2}\cdot 1.225\,kg/m^3\cdot 25\,cm^2\cdot 2.1\cdot 2a\Delta y$
$= \frac{1}{2}\cdot 1.225\,kg/m^3\cdot 25\times10^{-2}\,m^2\cdot 2.1\cdot 2\cdot 1.39\,m/s^2\cdot 1\,m$

$\boxed{\vec{D} = 8.94\times10^{-3}\,N}$

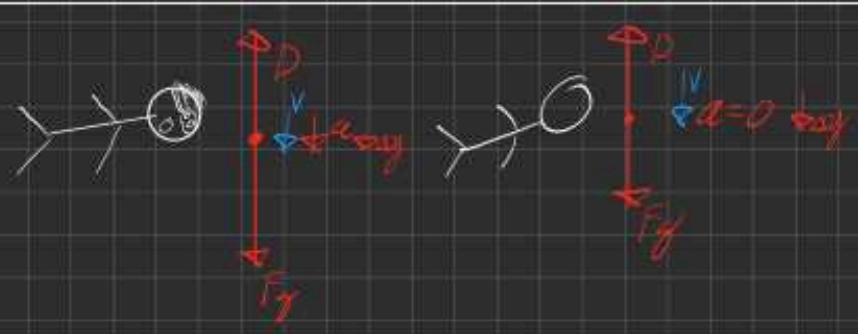
Assume D is insignificant then a = constant.

Uniformly Accel. Motion

$V = at + V_0$

$\Delta y = \frac{1}{2}at^2 + V_I t$

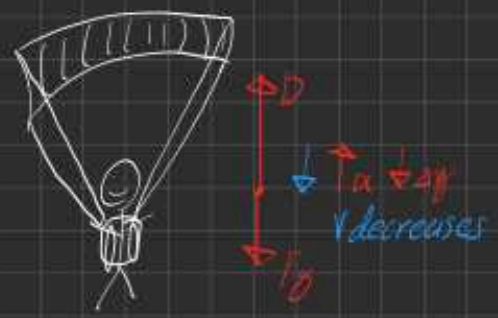
$\boxed{v^2} = V_0^2 + 2a\Delta y$



aerodynamic Lift
hydrodynamic Lift

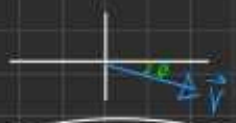
When a fluid/gas is turned by an obj

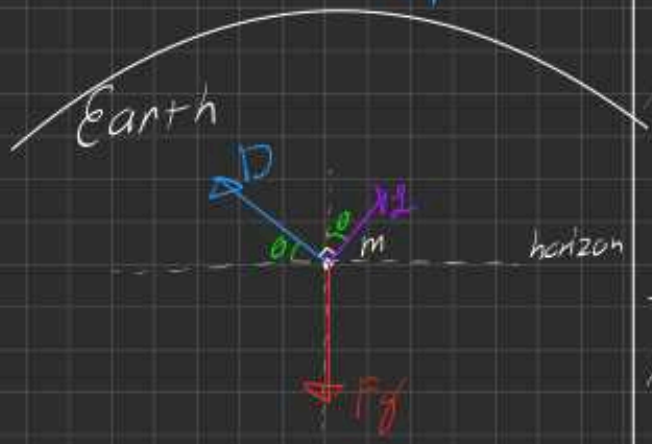
Movie?



Glide Ratio: Distance a glider travels horizont. over the dist. traveled vert.

$$\frac{\Delta x}{\Delta y} \propto \frac{L}{D}$$





Use Newt. 2nd Law to show glide ratio $\frac{L}{D} = \cot\theta$

or

$\frac{D}{L} = \tan\theta$

$F = ma$

$G.R. = \frac{hor.}{vert.}$

$F = ma$

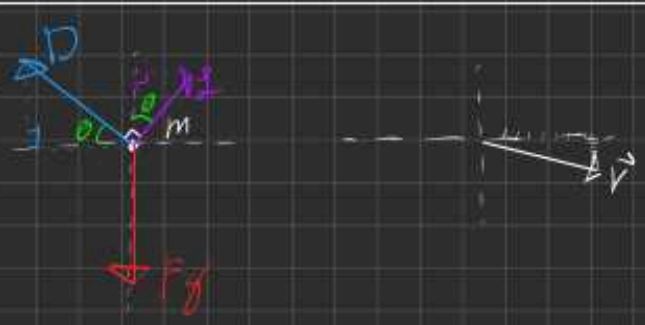
$(F_g + L + D + \vec{V}) = m(a\hat{i} + a\hat{j})$

$\vec{L} = [L\cos(90-\theta), L\sin(90-\theta)]$

$\vec{D} = [D\cos\theta, D\sin(180-\theta)]$

$= \cot\theta$

Ans:



If no wind & $L\sin\theta = D\cos\theta$
Dynamic Equilibrium $a=0$

$\cos 270°$
or

$\sum F = \hat{x}(L\sin\theta - D\cos\theta) + \hat{y}(L\cos\theta + D\sin\theta - mg) = 0$

$\frac{L}{D} = \cot\theta$

Obtain an expression for drag that doesn't include Lift.

$a = 0$

$F = ma$

$\hat{x} = 0 \quad \hat{y} = 0$

$\sum F = \hat{x}(L\sin\theta - D\cos\theta) + \hat{y}(L\cos\theta + D\sin\theta - mg) = 0$

$D\cot\theta\cos\theta + D\sin\theta - mg = 0$

$D(\cot\theta\cos\theta + \sin\theta) = mg$

$D = \frac{mg}{\cot\theta\cos\theta + \sin\theta} = \frac{1}{2}\rho C A V_t^2$

constant

terminal

$\boxed{\frac{L}{D} = \cot\theta \quad L = D\cot\theta}$

$L\sin\theta - D\cos\theta = L\cos\theta + D\sin\theta - mg$

$(-D\sin\theta - D\cos\theta = -mg)\,-1$

$D(\cos\theta + \sin\theta) = mg$

$D = \frac{mg}{\cos\theta + \sin\theta}$

Assume: $\theta = 21.8°$ $m = 65\,kg$ $\rho = 0.960\,kg/m^3$
$C = 1.2$ $A = 1.5\,m^2$

$$D = \frac{1}{2}\rho C A V_t^2 \qquad D = \left(\frac{65 \cdot 9.81}{\cot(21.8)\cos(21.8) + \sin(21.8)}\right) = 236.8N \approx 53\,lbs!!!$$

$$\frac{2D}{\rho C A} = V_t^2$$

$$\frac{2\left(\frac{65 \cdot 9.81}{\cot(21.8)\cos(21.8) + \sin(21.8)}\right)}{0.960 \cdot 1.2 \cdot 1.5} = V_t^2$$

$$\sqrt{274.07\,m^2/s^2} = \sqrt{V_t^2}$$

$$16.56\,m/s = V_t$$

$$\frac{16.56\,m}{s} \cdot \frac{60\,s}{1\,min} \cdot \frac{60\,min}{1\,hr} \cdot \frac{1\,mi}{1609.34\,m} \approx 37\,mph$$

Static aerostatic hydrostatic fluid

Side bar
Def 1
$m = \rho V$

Buoyant force

Equal to the weight of fluid displaced by an object. $\rightarrow F_g = mg$

$\boxed{B = m_f g}$ Def

$= \rho_f V_f g$

$V_{fluid\ displaced} = V_{object\ displacing\ fluid}$

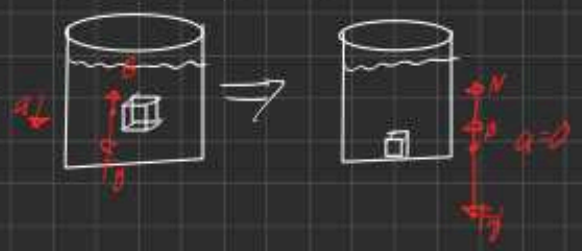
$\boxed{B = \rho_f V_{ob} g}$ Standard formula

$\rightarrow m_{obj} = \rho_{obj} \cdot V_{obj}$

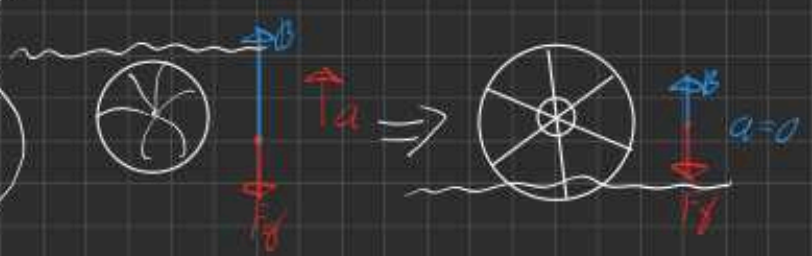
$V_{ob} = \frac{m_{obj}}{\rho_{obj}}$

$$B = \underbrace{\frac{\rho_f}{\rho_{obj}}}_{\text{Relative density = Specific gravity}} \cdot \underbrace{m_{obj} g}_{F_g = \text{weight}}$$

If $\rho_{obj} > \rho_f$



If $\rho_f > \rho_{obj}$



Human density 985 kg/m³

Fresh H_2O ρ 1000 kg/m³

Salt H_2O ρ 1025 kg/m³

Density vs Pressure

↳ Temperature

↳ Salinity

Oceans turning acidic

Don't drink deionized water & water

non-conductive

Gravity in space? Yes

Drag in space? No

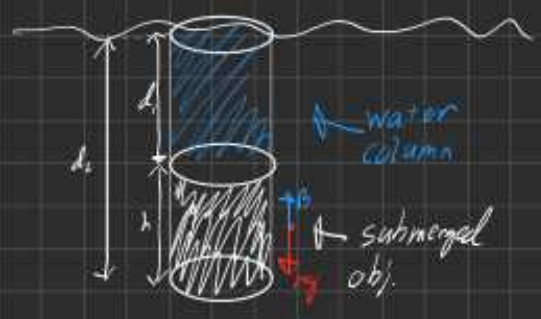
Boyancy in Space? No

Boyancy in deep ocean? Yes

Drag deep ocean? Yes

Density ρ_{H_2O} ~ constant surface to depth

Boyant B ~ nearly constant



$h = d_2 - d_1$

$B = \rho_f V_{obj}\, g$

$= \rho_f A \cdot h\, g$

$\rho_f \cdot A(d_2 - d_1) \cdot g$

$B = \rho_f d_2 g - \rho_f A d_1 g$

$\frac{F}{A} = \Delta P = \rho_f d_2 g - \rho_f d_1 g$

= Pressure bottom − Pressure Top

Thermohaline Circulation

0th Law of thermodynamics is that flow happens from hot to cold.

1) Warm surface water → polar regions (equatorial → polar)

2) Cools, Ice float, give salt, more dense, sinks

3) Gets replaced by warmer H_2O "drawn in"

4) Flows back to equator deep current.

5) Warm & rise as upwelling

Further Force Classifications

Restoring Forces

$F_{spr} = \text{Spring Force} = -k\Delta x$

spring displacement

spring constant

$f = \mu N$

later

Newton Definition V: Maintaining Circular Motion

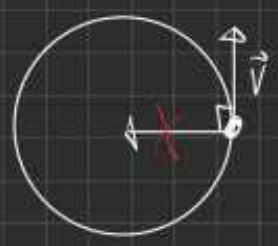
"A centripetal force is that by which bodies are drawn or implied, or any way tend, toward a point as to a centre."

Example 1: Orbits

"Of this sort is gravity, by which bodies tend to the center of the earth, ~~magnetism, by which iron tends to the loadstone;~~ and that force, what ever it is, by which the planets are perpetually drawn aside from the rectilinear motions, which otherwise they would pursue, and made to revolve in curvilinear orbitals."

Example 2: Sliing

"A stone, whirled about in a sling, endeavors to recede from the hand that turnes it, …"



Example 3: Projectile Motion

A projectile, if it was not for the force of gravity, would not deviate towards the earth, but would go off from it in a right line, and that with an uniform motion, if the resistance of the air was taken away. It is by its gravity that it is drawn towards the earth, more or less., according to the force of its gravity, and the velocity of its motion.

Definition VI:

"The absolute quantity of a centripetal force is the measure of the same, proportional to the efficacy of the cause that propagates it from the center through the spaces round about.

$W = F \cdot \Delta x$ $\qquad$ $r \cdot F_c = mv^2$

Torque

It is important to make a distinction between that which causes rotational motion, and that which maintains rotational motion.

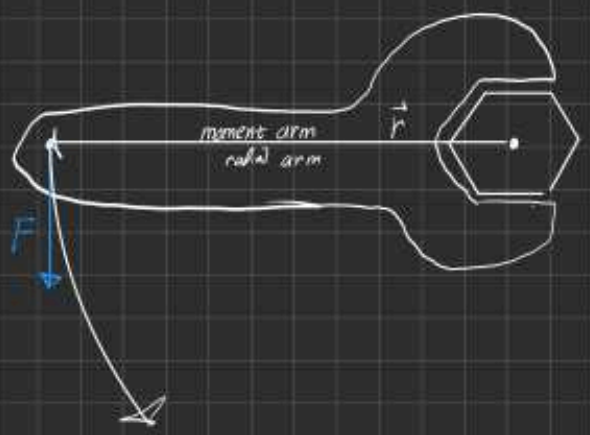
Torque = $\vec{\tau} = \vec{r} \times \vec{F}$

rotational analogue to Force

$= r F \sin\theta$

$r F \sin 0^\circ = 0$ $\quad \parallel$ vectors

$r F \sin 90^\circ =$ $\quad \perp$ vectors

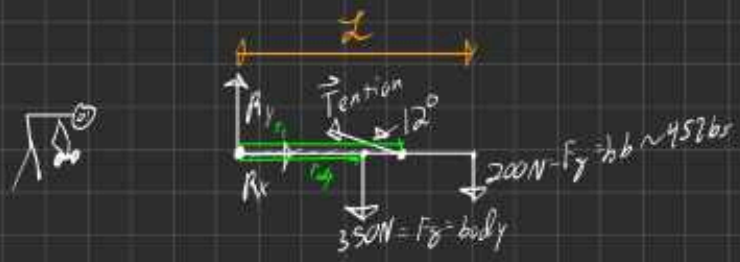


Equilibrium

Static: $\vec{v} = 0$, $a = 0$, No rotation!

Dynamic: $a = 0$, $v =$ constant, No rotation!

$\sum F = 0 \quad \sum \tau = 0$



$\sum \tau = \tau_T - \tau_{body} - \tau_{bb} = 0$

$r_T T \sin 12^\circ - r_{body} F_{body} \sin 90^\circ - r_{bb} F_{bb} \sin 90^\circ$

$\frac{2}{3} \qquad \frac{1}{2} \qquad 1$

$r_{body} F_{body} \cdot 1 + r_{bb} F_{bb} \cdot 1 = r_T T \sin 12^\circ$

$\frac{1}{2} \cdot 350N + 1 \cdot 200N = \frac{2}{3} T \sin 12^\circ$

2705 N = T

① F-body diagram

② Motion?

③ $\sum F \quad \sum \tau$

compressional force

$\sum F = R_x - T_x = 0$

1262

$R_x = T \cos 12^\circ$

body w/ bb: $R_x = 6082 lbs \cdot \cos 12$, $R_x = 5952 lbs$

body: $R_x = 2842 lbs \cdot \cos 12$, $R_x = 2782 lbs$

~1442

$2705N \cdot \frac{1 lb}{4.484 N} = 608 lbs$ body + bb

$1263N \longrightarrow 284 lbs$ body

Lifting bb at this position would add 324 lbs

Summary of Moment Orders

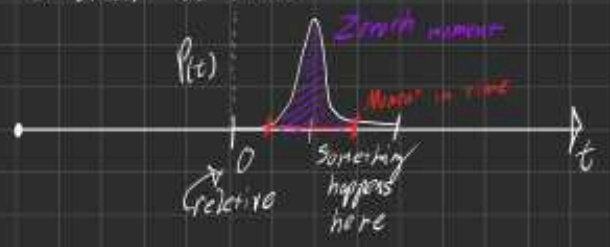
Moment (Order)	Statistical Meaning	Physical Interpretation	Example
0th	Total probability (normalization)	Conservation / total quantity	100% of students eventually leave
1st	Mean / expected value	Fulcrum, Center, Average	Average time students leave
2nd	Variance / spread	Dispersion, width of distribution	Range of early vs. late departures
3rd	Skewness	Asymmetry of distribution	More students leave early vs. late
4th	Kurtosis	Extremes / tail heaviness	Sharp rush or slow filter-out pattern

Summary of Moment Orders for Mass

Mass Moment	Definition (typical)		Meaning	Units
0th	$M_0 = \int \rho \, dV$		Total mass	kg
1st (Raw, Vector)	$M_1 = \int \rho \vec{r} \, dV$		Center of mass via $\vec{R} = \vec{M}_1 / M_0$	kg m
2nd	$M_2 = I_O = \sum_i m_i \lVert \vec{r}_i \rVert^2$		Moment of inertia (rotation)	kg m^2
3rd (along x)	$\mu_3 = \int (x - \bar{x})^3 \lambda(x) \, dx$	with linear density λ	Mass asymmetry about the mean	kg m^3
4th (along x)	$\mu_4 = \int (x - \bar{x})^4 \lambda(x) \, dx$	with linear density λ	Concentration of mass & mass outliers	kg m^4

Stats Review

Moment of time

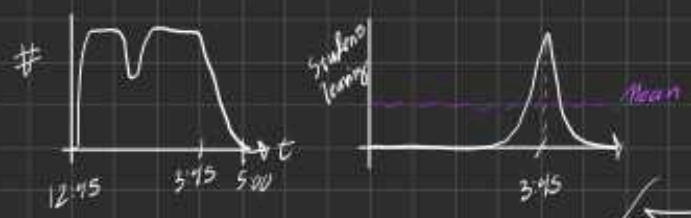


Zeroth Moment (μ_0): Total Probability $= \int P(t)\,dt$

- we're sure something is going to happen
- Normalization: 1 or 100% is maximum
  ↳ usual case

First Moment: Mean $\bar{t} = \mu_1 = \int t\,P(t)\,dt$

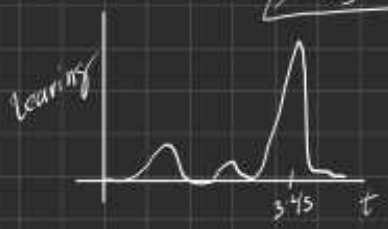
- Average or Expectation



Normalized $\bar{t} = \frac{\mu_1}{\mu_0} = \frac{\int t\,P(t)\,dt}{\int P(t)\,dt}$

Second moment: Scattering/variance

$$\mu_2 = \int (t-\bar{t})^2 P(t)\,dt$$



Third Moment: Imbalance - left or right skewed

$$\mu_3 = \int (t-\bar{t})^3 P(t)\,dt$$

Fourth Moment: Kurtosis - peak vs spread

- Mesokurtic: Gaussian Standard
- Leptokurtic: Strong peak, heavy tails
- Platykurtic: weak peaks, light tails

Not Quizzed on these moments

Law of the Lever

Magnitudes are in equilibrium at distances reciprocally proportional to their weights.

Consider a mean of grades 48 and 42 out of 50

$48 + 42 = 90 \div 2 = 45$

Normalize: $\frac{45}{50} \cdot 100 = 90\%$

What if different weights

48 @ 60% & 45 @ 40%

$48 \cdot 0.6$ $\quad$ $45 \cdot 0.4$

28.8 $+$ 18

$\frac{468}{50} \cdot 100 = 93.6$

Weighted Avg: $\bar{x} = \frac{\sum_i x_i w_i}{\sum_i w_i}$ ← Weighted Avg ← Normalization

If $m_1 = m_2$: Equilibrium (static)

$m_1 > m_2$: m_1 rotates down
m_2 rotates up

Suppose $m_1 = 30$ kg & $m_2 = 5$ kg

Suppose $m_1 = 30\,kg$ & $m_2 = 5\,kg$

Utilize the 1 spatial moment of mass as the weighted average to find r, where to put the fulcrum.

zeroth moment of mass: Total mass of system
first moment of mass: center of mass.

$$\vec{R} = \frac{\sum_i m_i r_i}{\sum_i m_i} = \frac{30\,kg(-2)+5(2)}{30\,kg+5\,kg} = \frac{-60\,kg+10\,kg}{35\,kg} = -1.43$$

What about a plane

$$M_0 = \sum_i m_i = 6+2+1+4 = 13$$

$$\bar{x} = \frac{M_y}{M_0} = \frac{3}{13} = 0.23$$

$$\bar{y} = \frac{M_x}{M_0} = \frac{1}{13} = 0.077$$

Center of mass
$(0.23, 0.077)$

Fulcrum shifts right & up

$$M_y = \sum_i m_i x_i = 6\cdot 1+2\cdot 1-1\cdot 1-4\cdot 1 = 3$$

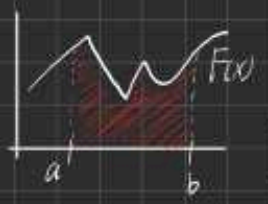
$$M_x = \sum_i m_i y_i = 1+6-4-2 = 1$$

Continuous Mass

$\rho = \frac{M}{V}$

sigma $\sigma = \frac{M}{A}$

lambda $\lambda = \frac{M}{L}$



1/2 f(x) gives balance about x

$M_0 = \sum_i M_i$ but $m = \sigma A$ where $A = f(x)dx$

$= \sum_i^N \sigma f(x_i)dx$ as N gets larger, take limit of the sum to get integral

$= \frac{1}{n}\lim_{n\to\infty}\sum_{i=1}^{N} \sigma f(x_i)dx$

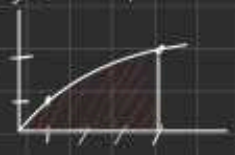
$= \sigma\int_a^b f(x)dx$

$M_x = \sigma\int_a^b y f(x)dx$

$\sigma\int_a^b \left(\frac{1}{2}f(x)\right)f(x)dx = \frac{1}{2}\sigma\int_a^b f(x)^2 dx$

$M_y = \sigma\int_a^b x f(x)dx$

For $f(x) = \sqrt{x}$ from $0 \to 4$



$M_0 = \sigma\int_a^b f(x)dx$

$= \sigma\int_0^4 x^{1/2}dx$

$= \sigma\left[\frac{2}{3}x^{3/2}\right]_0^4$

$= \sigma\frac{16}{3}$

$M_y = \int_0^4 x\,\sigma(\sqrt{x})dx$

$= \sigma\int_0^4 x\sqrt{x}\,dx$

$= \sigma\int_0^4 x^{3/2}dx$

$= \sigma\left[\frac{2}{5}x^{5/2}\right]_0^4$

$= \sigma\frac{64}{5}$

$M_x = \frac{1}{2}\int_0^4 \sigma(\sqrt{x}^2)dx$

$= \frac{1}{2}\sigma\int_0^4 x\,dx$

$= \frac{1}{2}\sigma\left[\frac{x^2}{2}\right]_0^4$

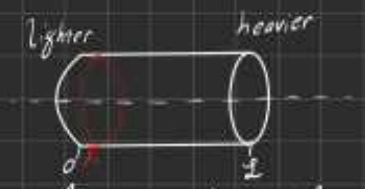
$= \sigma\frac{1}{4}[4^2-0]$

$= \sigma 4$

$\bar{x} = \frac{M_y}{M_0} = \frac{\sigma 32}{\sigma\frac{16}{3}} = \frac{12}{5} = 2.4$ | $\bar{y} = \frac{M_x}{M_0} = \frac{\sigma 4}{\sigma\frac{16}{3}} = 0.75$

Center of mass: $(2.4, 0.75)$

$\vec{R} = [2.4, 1.5]$



$\rho = \frac{M}{V}$ $\quad m = \rho V$

$dm = \rho dV$

$= \rho dA dl$

$= \rho A dl$ Nice way

$\rho = \rho \cdot l$ (constant; ρ_{new} increases in mass)

$dm = \rho l A dl$

l

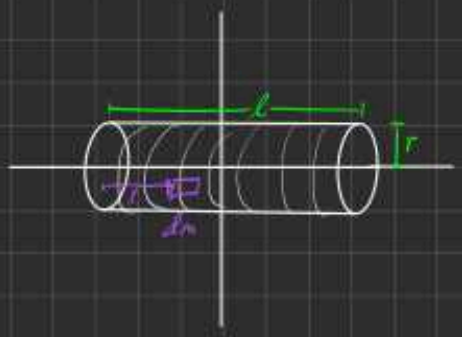
$l\,l = A$

$l\,l\,l = V$

Center of mass on Exam & Final

Try to calc moments of inertia for diff types.

$$M_0 = \int_0^L \rho \, dV$$
$$= \int_0^L (C l)(A \, dl)$$
$$= C A \int_0^L l \, dl$$
$$= C A \left[\frac{l^2}{2}\right]_0^L$$
$$= C \cdot A \cdot \frac{L^2}{2}$$
$$M_0 = \frac{1}{2} C \cdot A \cdot L^2$$



$$M_x = \int dm \cdot r$$
$$= \int (C l A \, dl) \cdot r$$
$$= C A r \int l \, dl$$
$$= C A r \left[\frac{l^2}{2}\right]_0^L$$
$$M_x = \frac{1}{2} C A r L^2$$

$$M_y = \int dm \cdot l$$
$$= \int (C l A \, dl) \cdot l$$
$$= C A \int l^2 \, dl$$
$$= C A \left[\frac{l^3}{3}\right]_0^L$$
$$M_y = \frac{1}{3} C A L^3$$

$$\bar{x} = \frac{M_y}{M_0} = \frac{\frac{1}{3} C A L^3}{\frac{1}{2} C A L^2}$$
$$\bar{x} = \frac{3}{2} L$$

$$\bar{y} = \frac{M_x}{M_0}$$
$$\bar{y} = \frac{\frac{1}{2} C A r L^2}{\frac{1}{2} C A L^2}$$
$$\bar{y} = r$$

Center of mass $(\frac{3}{2} L, r)$

Zeroth moment
total mass

First moment
normalized
center of mass

Second moment
Moment of Inertia (I): $m \cdot l^2$
$I = \sum_i m_i r_i^2$ discrete mass

If we rotate the cylinder about O

$$dI = dm \, l^2$$
$$= \rho \, dV \, l^2$$
$$\int dI = \int (C l)(A \, dl) l^2$$
$$I = C A \int l^3 \, dl$$
$$I = C A \left[\frac{l^4}{4}\right]_0^L$$
$$I = \frac{1}{4} C A \cdot L^4$$
$$= \frac{1}{2}\left(\frac{1}{2} C A L^2\right) L^2$$
$$I = \frac{1}{2} M_0 L^2$$

$$M_0 = \rho V = \frac{1}{2} C A L^2$$

moment of Inertia

Non-Uniform
rod

For Uniform rod : $\frac{1}{3} M L^2$

Inertial moment of about

$f(x)$ & $g(x)$

$$M_0 = \rho \int_a^b f(x) - g(x) \, dx$$
$$M_x = \frac{1}{2} \rho \int [f(x)^2 - [g(x)]^2] \, dx$$
$$M_y = \rho \int x [f(x) - g(x)] \, dx$$
$$\bar{x} = \frac{M_y}{M_0} \quad \bar{y} = \frac{M_x}{M_0}$$

2nd $M_{2x} = \frac{1}{3} \rho \frac{1}{3} [$

Summary of Motion Moment Orders for Mass

Mass Motion Moment	Definition (typical)	Meaning	Units
0th	$M_0 = \int \rho \, dV$	Total mass	kg
1st (Raw, Vector)	$M_1^{(v)} = P = MV$	Momentum of mass	kg m/s
2nd	$M_2^{(v)} = \sum_i m_i \, \lVert \vec{v}_i \rVert^2$	Virial relation: proportional to time-averaged kinetic energy; connects motion and binding forces in steady-state systems.	kg m^2/s^2 = Nm = J

Motion Moments of Mass:

Zeroth: Total Mass: $M_0 = \sum_i M_i$

First: $M_1^{(\vec{v})} = \sum_i M_i \cdot \vec{V}_i = \vec{p}$ ← momentum

If $\vec{r}$ is the vector from (0) to center of mass (com)

then $\vec{V} = \frac{dr}{dt} = \frac{d}{dt}\left[\frac{1}{M_0}\sum_i M_i r_i\right]$

$$= \frac{1}{M_0}\sum_i M_i \frac{d\vec{r}_i}{dt} = \underline{\underline{\frac{1}{M_0}\sum_i M_i \vec{V}_i}}$$

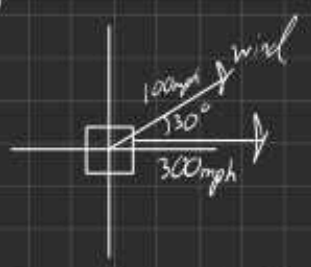
Momentum of com = $m \cdot V_{(com)}$ $\quad \vec{p} = m_0 \vec{V}_{com}$

Second Motion Moment

$M_2^{(\vec{v})} = \sum_i m_i (\vec{V}^2)$ $\quad$ Time avg kinetic Energy

$\|V^2\|$

Airplanes - Relative Velocity



$$\sum F = (300 + 100\cos(30°))\hat{x} + (0 + 100\sin(30°))\hat{y}$$
$$= 387\text{mph}\,\hat{x} + 50\text{mph}\,\hat{y}$$

$h = \sqrt{(387)^2 + (50)^2}$ $\quad \tan\theta = \frac{o}{a}$

$h = 390\text{mph}$ $\quad \theta = \tan^{-1}\left(\frac{50}{387}\right)$

$\theta = 7.36°$

wind Correction: 7.37° South of East

✓ Spatial

Spacial Moments of Force

0th: Net Force $= \frac{d\vec{p}}{dt}$ ← momentum

$$= \frac{d}{dt}\left[\sum_i m_i \vec{V}_i\right]$$

$$= \sum_i m_i \frac{d\vec{V}_i}{dt} = \sum_i m_i \vec{a}_i = M_0 \vec{a}_{com}$$

---

w/ current 2mi in 40min $\quad a = 0$

against current 2mi in 60min $\quad$ dynamic equilibrium

$V_l = \frac{2mi}{40min}$ $\quad V_u = \frac{2mi}{60min}$

$\vec{V}_l = \vec{V}_s + \vec{V}_c$ $\quad -\vec{V}_u = -\vec{V}_s + \vec{V}_c$

$\vec{V}_l - \vec{V}_s = \vec{V}_c$ $\quad \vec{V}_s - \vec{V}_u = \vec{V}_c$

$$\vec{V}_l = \vec{V}_s + \vec{V}_c$$
$$-(-\vec{V}_u = -\vec{V}_s + \vec{V}_c)$$
$$\vec{V}_l + \vec{V}_u = 2\vec{V}_s$$
$$\frac{\vec{V}_l + \vec{V}_u}{2} = \vec{V}_s$$
$$\frac{\frac{2}{40} + \frac{2}{60}}{2} = \vec{V}_s$$
$$\frac{1mi}{40min} + \frac{1mi}{60min} = \vec{V}_s$$
$$\frac{3mi}{120min} + \frac{2mi}{120min} = \vec{V}_s$$
$$\frac{5mi}{120min} = \vec{V}_s$$
$$\frac{1mi}{24min} \cdot \frac{60min}{1hr}$$
$$2.5mi/hr = \vec{V}_s$$

down	up	still
$\frac{1mi}{20min}$	$\frac{1mi}{30min}$	$\frac{1mi}{24min}$

Spatial Moments of Force

Quantity	Moment	Definition (typical)	Meaning	Units	Standard Units
Force	0th	$\vec{F}_{net} = M\vec{A}_{cm}$	Net Force	N	N
Work	1st (Scalar)	$W = \int_C \vec{F}_{net} \cdot d\vec{r}_{cm}$	Energy transfer by force through displacement	Nm = J	J
Virial Relation	1st (Scalar)	$m\,\|\|v^2\|\| = \vec{r}_{cm} \cdot \vec{F}_{net} = \vec{r}_{cm} \cdot \vec{F}_c$	Connects work & energy, and leads to centripetal force	Nm = J	Nm
Torque	1st (Vector)	$\vec{\tau}_O = \vec{r}_{cm} \times \vec{F}_{net}$	Rotational force about point/axis	Nm = J	Nm

Spatial Moments of Force

Quantity	Moment	Definition (typical)	Meaning	Units	Standard Units
Force	0th	$\vec{F}_{net} = M\vec{A}_{cm}$	Net Force	N	N
Power	1st (Scalar)	$P = \vec{F}\cdot\vec{V} = \frac{dW}{dt}$	Scalar moment — Instantaneous Power; rate of work or energy transfer	Nm/s = J/s	J/s
	1st (Vector)	$M_{P1}^{(\vec{v})} = \vec{V}\times\vec{F}_{net}$	Vector moment — deflection of mass		Nm/s (likely)

1st Moment of Force (spatial): $\vec{r} * \vec{F}$

a) Translational (dot)

"quasi-First moment"

$$\text{Work} = \int \vec{F}\cdot d\vec{r}$$

Scalar; Force (vector); displacement

1) Find Net force then multiply by displacement
2) Find Work done by each force, then add

b) Rotational "first moment of Force"

1) $\vec{r}\cdot\vec{F}$ (Nm) 2) $\vec{r}\times\vec{F} = \tau$ (Nm)

$\vec{r}\cdot F$

① $rF_t = mv_T^2$

$r\times F = \tau$ $r\times ma$

$r\times m(r\alpha)$

$mr^2\alpha = \tau$

$I\alpha = \tau$

Newton 2nd Law for Rotations

Gravitron!

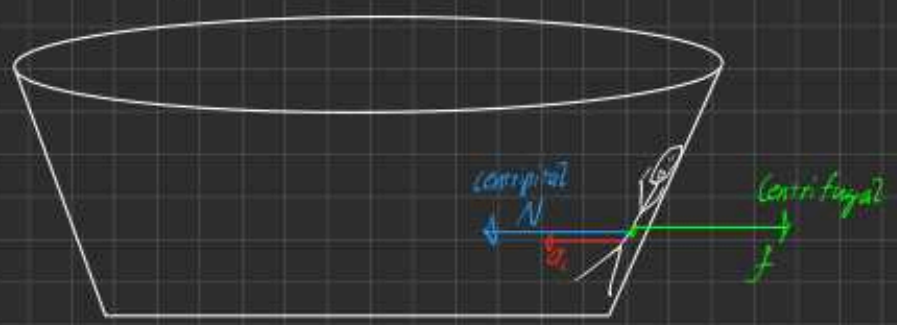
This carnival ride moves orders in an enclosed pod to 24rpm in 20s. Calculate angular speed at 20s.

24rpm → rad/s Diameter = 30ft

$$\vec{\omega} = \frac{24\,rot}{min}\cdot\frac{1\,min}{60s}\cdot\frac{2\pi\,rad}{1\,rot} = 2.51\,s^{-1}$$

$$\vec{V} = r\omega = 4.57m\cdot 2.51s^{-1} = 11.6\,m/s\cdot\frac{2.24\,mph}{\frac{m}{s}} = 26\,mph$$

$$15ft\cdot\frac{0.3047}{1ft} = 4.57m$$



$\vec{a}_c = r\omega^2$ $\vec{V} = r\omega$

$= r\frac{V^2}{r^2}$

$a_c = \frac{V^2}{r}$

$m = 63.5\,slug$

momentum = $\vec{p}$

$F_c = N = ma_c = mr\omega^2$

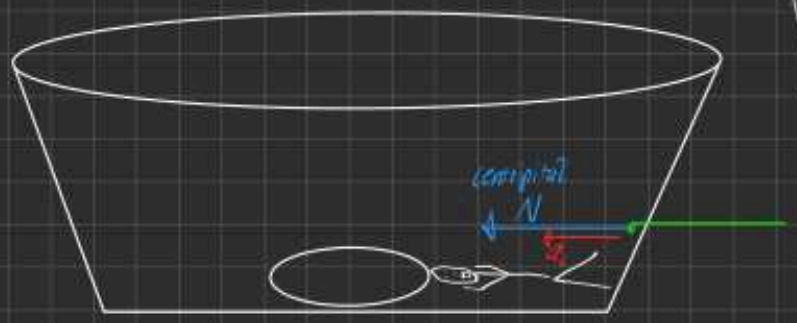
$= 63.5slug\cdot 4.57m\cdot(2.51s^{-1})^2$

$= 1872.22N\cdot\frac{1lb}{4.448N} = 421\,lbs$!!!

$p = \frac{F}{A}$

$psi = \frac{lb}{in^2} \rightarrow \frac{F}{A}$

$\Delta\theta \approx 3\theta$



$a_c = \frac{v^2}{r}$ $\quad v = r\omega$

Calculate average angular acceleration

$a = r\alpha$

$r = \frac{9.14m}{2}$

$\omega_0 = 0$
$\omega_f = 2.54 \frac{rad}{s}$
$t = 20s$

$\alpha =$
$\Delta\theta =$

$\alpha = constant$

$\omega = \alpha t + \omega_0$

$\Delta\theta = \frac{1}{2}\alpha t^2 + \omega_0 t$

$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$

$2.54\ rad/s \cdot 20s$
$\alpha = 0.127\ rad/s^2$

$a = r\alpha$
$= 4.57 \cdot 0.127$
$= 0.58$

$\Delta\theta = \frac{1}{2}(0.126 \frac{rad}{s^2})(20s)^2$

$\Delta\theta = 252\ rad \cdot \frac{1\ rev}{2\pi\ rad}$

$\Delta\theta = 4\ rev$

Motion Moments of Force: Some common, some not, but useful!

0th: Net Force

1st: $\vec{v} \cdot \vec{F}$ - - - - - - - - - ξ - - - $\vec{v} \times \vec{F}$
Rotational Power

$\downarrow$

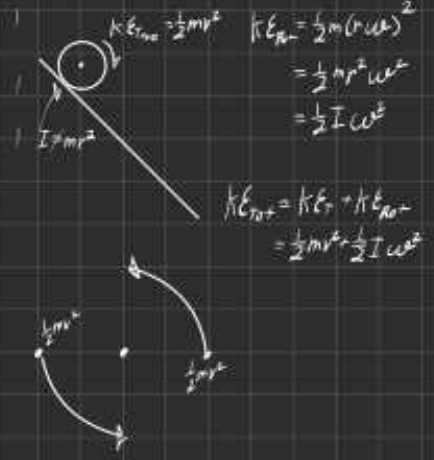
$\underbrace{\frac{dR_{cm}}{dt} \cdot F}$

Power $= \frac{dW}{dt} \leftarrow \frac{\text{work done}}{\text{unit time}} = \frac{\Delta KE}{\Delta t}$

W = Energy Theorem

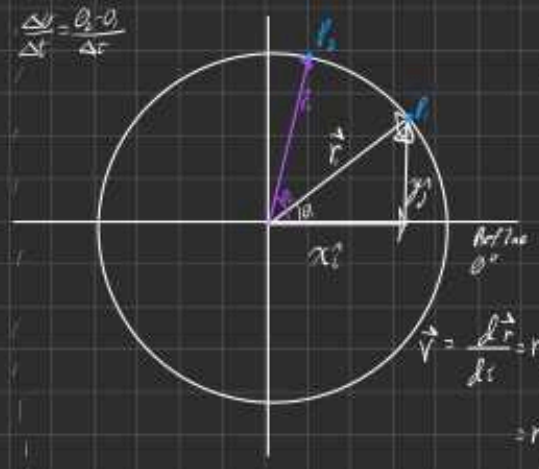
$W = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_0^2$

Time moment of force: $\int \vec{F} \cdot dt$



Projectile

$\frac{\Delta\omega}{\Delta t} = \frac{\omega_2 - \omega_1}{\Delta t}$



$r\hat{r} = x\hat{i} + y\hat{j}$

$= r\cos\theta\hat{i} + r\sin\theta\hat{j}$

$= r(\cos\theta\hat{i} + \sin\theta\hat{j})$

$\vec{V} = \frac{d\vec{r}}{dt} = r\frac{d}{dt}\left[\cos\theta\hat{i} + \sin\theta\hat{j}\right]$

$= r\left[-\sin\theta\frac{d\theta}{dt}\hat{i} + \cos\theta\frac{d\theta}{dt}\hat{j}\right]$

$= r\frac{d\theta}{dt}\left[-\sin\theta\hat{i} + \cos\theta\hat{j}\right]$

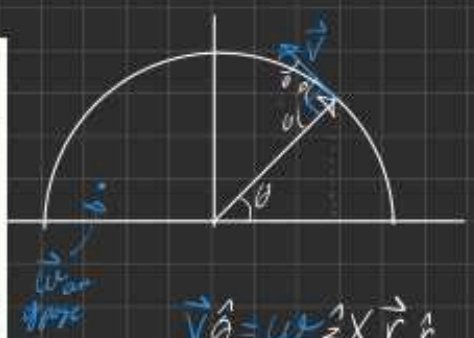
Angular Velocity (ω)

Angular position $= \theta$

Angular displacement $= \Delta\theta$

Angular Velocity (ω)

$\vec{V} = r\omega\left[-\sin\theta\hat{i} + \cos\theta\hat{j}\right]$



$\frac{d\vec{r}}{dt} \approx \frac{ds}{dt} = r\frac{d\theta}{dt}$

$ds = r\,d\theta$

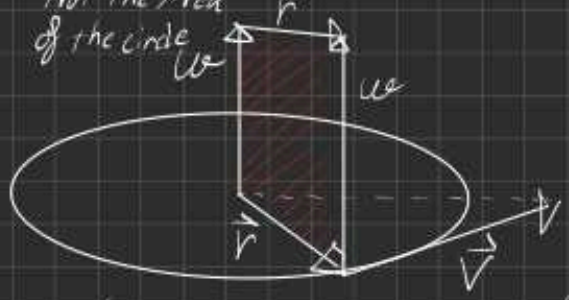
$s = r\theta$

Kinematics for Projectile motion.

	Uniform Motion	Uniformly Accelerated Motion
	x	y
Position	$x = v_{0x}t + x_0$ $= v_0 \cos\theta\, t + x_0$	$y = -\frac{1}{2}gt^2 + v_{0y}t + y_0$ $= -\frac{1}{2}gt^2 + v_0 \sin\theta\, t + y_0$
Velocity	$v_x = v_{0x} = v_0 \cos\theta$	$v_y = -gt + v_{0y}$ $= -gt + v_0 \sin\theta$
Acceleration	0	$a_y = a_{0y} = -g = -9.81\,m/s^2$
Misc.		$v_y^2 = (v_0 \sin\theta)^2 - 2g\Delta y$
Free-Fall?	NOT!!!	Free-Fall!!!

$\vec{V}\hat{\theta} = \omega\hat{z} \times \vec{r}\,\hat{r}$

always '+' (curvilinear)



By using polar we go from 2 var. x, y to 1 var & 1 const θ, r

$a_t = \alpha = 0$

No $\mathcal{F}$ only $F_{centripetal}$

Uniform circular motion

If $\vec{V} \equiv$ constant $\equiv \omega$

$\vec{V} = r\omega\left[-\sin\theta\hat{i} + \cos\theta\hat{j}\right]$

$\vec{V} = r\left[-\omega\sin\theta\hat{i} + \omega\cos\theta\hat{j}\right]$

$\frac{\Delta\theta}{\Delta t} = \omega$ $\qquad \Delta\theta = \omega\Delta t$

$\frac{d\theta}{dt} = \vec{\omega}$ $\qquad d\theta = \omega\,dt$

$\vec{a} = \frac{d\vec{V}}{dt} = r\frac{d}{dt}\left[-\omega\sin\theta\hat{i} + \omega\cos\theta\hat{j}\right]$

$= r\left[\frac{d}{dt}(-\omega\sin\theta\hat{i}) + \frac{d}{dt}(\omega\cos\theta\hat{j})\right]$

$= r\left[\left(-\frac{d\omega}{dt}\sin\theta - \omega\cos\theta\frac{d\theta}{dt}\right)\hat{i} + \left(\frac{d\omega}{dt}\cos\theta - \omega\sin\theta\frac{d\theta}{dt}\right)\hat{j}\right]$

α $\qquad \alpha$

Angular acceleration

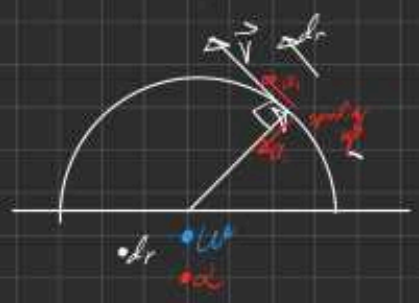
$\vec{a} = \underbrace{r\omega^2\left[-\cos\theta\hat{i} - \sin\theta\hat{j}\right]}_{QIII} + \underbrace{r\alpha\left[-\sin\theta\hat{i} + \cos\theta\hat{j}\right]}_{QII}$

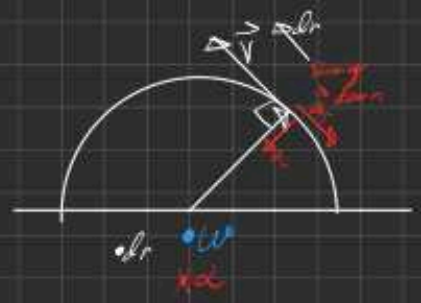
a_c Centripetal $\qquad a_t$ Tangential

$\vec{V} = r\omega\left[-\sin\theta\hat{i} + \cos\theta\hat{j}\right]$

$\vec{V} = r\left[-\omega\sin\theta\hat{i} + \omega\cos\theta\hat{j}\right]$

$\cancel{\vec{V} = r\left[-\omega\sin\theta + \omega\cos\theta\right]} \quad \cancel{\vec{V} = r\left[-\frac{d\theta}{dt}\sin\theta + \frac{d\theta}{dt}\cos\theta\right]} \quad \cancel{\vec{V} = r\left[-\sin\theta - \frac{d\theta}{dt}\cos\theta\right]}$





Revolution of earth

$$\omega_{rev} = \frac{1\,rev}{1\,yr} \cdot \frac{1\,yr}{365.25\,day} \cdot \frac{1\,day}{24\,hr} \cdot \frac{1\,hr}{60^2\,s} \cdot \frac{2\pi\,rad}{1\,rev} = 2 \cdot 10^{-8}\,rad/sec$$

$$\omega_{rot} = \frac{1\,rev}{1\,day} \cdot \frac{1\,day}{24\,hr} \cdot \frac{1\,hr}{60^2\,s} \cdot \frac{2\pi\,rad}{1\,rev} = 7.27 \cdot 10^{-5}\,\frac{rad}{sec}$$

$$= \frac{1\,rev}{24\,hr} \cdot \frac{1\,hr}{60\,min} = 6.9 \cdot 10^{-4}\,rpm$$

$$\vec{V} = r\omega = 6371 \cdot 10^3\,m \cdot 7.27 \cdot 10^{-5}\,s^{-1} \cdot \frac{2.24\,mph}{\frac{m}{s}} \approx 1000\,mph$$

$1\,rev = 2\pi\,rad = 360^\circ$

SI

total arc length $\leftarrow$ $\overset{(m)}{\Delta S} = \overset{(m)}{r}\,\overset{(rad)}{\Delta\theta}$ $\leftarrow$ angular distance

radius or rotation

$$\Delta\theta = \frac{\Delta s\,m}{r\,m}$$

radians is dimensionless

Angular position $\equiv$ rad

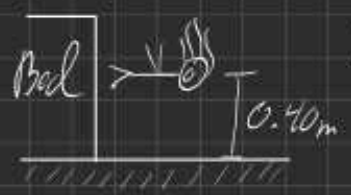
ω Velocity $= \frac{d\omega}{dt} = \frac{rad}{s} = \frac{1}{s} = s^{-1}$

α acceleration $= \frac{d\omega}{dt} = \frac{rad}{s^2} = \frac{1}{s^2} = s^{-2}$

Traumatic Brain Injury

Paul-Leah B-day
≈ January

If $a > m/s^2 > 1ms$ ≈ 100G
TBI can occur



Acceleration
$\Delta y \downarrow \; v \downarrow \; a_g \downarrow$

Deceleration
$\Delta y \downarrow \; v \downarrow \; a_g \uparrow$
(2mm hardwood
(1cm carpet

$\underbrace{V_0 = 0 \rightarrow V_f = V_I}_{a_g}$

$\underbrace{V_0 = V_I \rightarrow V_f = 0}_{\text{Impact}}$

$a = ? \rightarrow TBI$

$\Delta y = -0.40m$

$V_0 = 0$

$a = -9.81 m/s^2$

$V_f = V_I$

$t = ?$
time independant

$\Delta y \begin{cases} 2mm \\ 1cm \end{cases}$

$V_f = 0$

$V_0 = V_I = ?$

$\boxed{t = ?}$

$\boxed{a = ?}$

$V_I^2 = V_0^2 + 2g\Delta y$

$V_I = \pm\sqrt{2g\Delta y}$

$\sqrt{2 \cdot 9.81 \cdot 0.4}$
$\sqrt{7.85}$

$V_I = 2.8 m/s, \; \frac{2.24mi}{1m/s} = 6.28mph \rightarrow V_0 = V_I \rightarrow V_f = 0$

Impact
$a \equiv TBI?$

Work Fall

if $m_{adult} = 5kg$

$W_g = F_g \cdot \Delta y_{fall}$

$m_{child} = 5 \cdot 0.25 = 1.25 kg$

$= m \cdot 9.81 \frac{m}{s^2} \cdot 0.4m$

$W_g = 1.25kg \cdot 9.81 \frac{m}{s^2} \cdot 0.4m$

$W_g = 4.9 \frac{kgm^2}{s^2}$ or $4.9Nm$ or $4.9J$

Work Impact

HW: $W = 1.25kg \cdot 1960 m/s^2 \cdot (-2 \cdot 10^{-3})m$
$W = -4.9J$

Carpet: $W = 1.25kg \cdot 392 m/s^2 \cdot (-1 \cdot 10^{-2})m$
$W = -4.9J$

$\Delta y \begin{cases} a \; 2mm \rightarrow 0.002m \\ b \; 1cm \rightarrow 10mm \rightarrow 0.01m \end{cases}$

$V_f = 0$

$V_0 = V_I = 2.8 m/s$

$\boxed{t = ?}$

$\boxed{a = ?}$

$V^2 = V_0^2 + 2a\Delta y$

a
$V_0^2 = 2a\Delta x$
$a = \frac{(2.8m/s)^2}{2 \cdot 0.002m}$
$a = 1960 m/s^2$
$\approx 2000 m/s^2$
Hardwood

b
$V_0^2 = 2a\Delta x$
$a = \frac{(2.8m/s)^2}{2 \cdot 0.01}$
$a = 393 m/s^2$
$\approx 400 m/s^2$
Carpet

$V_f = at + V_0$

$t = -\frac{V_0}{a}$

$t = -\frac{(-2.8m/s)}{2000m/s^2}$

$t = 0.0014 sec, \; \frac{1000ms}{1sec} = 1.4ms$

HW

TBI For sure!!!

~7ms
Carpet

Dollar Bill

15.50cm ≡ Length of a $bill

7.75cm = 0.0775m

$V_f^2 = V_0^2 + 2a\Delta y$

$a = \pm\sqrt{\frac{2(0.0775m)}{9.81m/s^2}}$

$a = 0.126$

$\Delta y = 0.0775m$

$V_0 = 0$

$V_f = ?$ NO

$a = -9.81 m/s^2$

$* t = ?$

$\Delta y = -\frac{1}{2}gt^2 + \cancel{V_0 t}$

$\cancel{V_f = -gt + V_0}$

$t^2 = -\frac{\Delta y \cdot 2}{g}$

$t = \pm\sqrt{-\frac{(-0.0775m)\cdot 2}{9.81m/s^2}}$

$\boxed{t = 0.126s}$

types of motion

$x(t) = \underbrace{\frac{1}{3}t^3}_{\text{cube}} + \underbrace{4t^2}_{\text{Uni accel}} + \underbrace{15t}_{\text{Uni}} - \underbrace{10}_{\text{Const}}$

Derive all kinematic equations for this motion:

$V(t) = t^2 - 8t + 15$

$a(t) = 2t - 8$

$J(t) = 2$

From t=0 to t=6 when is it at rest

$0 = t^2 - 8t + 15$

$0 = (t-5)(t-3)$

$0 = t-5 \quad 0 = t-3$

$t = 5 \quad t = 3$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-(-8) \pm \sqrt{(8)^2 - (4 \cdot 15)}}{2}$

$= \frac{8 \pm \sqrt{64 - 60}}{2}$

$= \frac{8 \pm \sqrt{4}}{2}$

$= \frac{8-2}{2} \quad = \frac{8+2}{2}$

$= \frac{6}{2} \quad = \frac{10}{2}$

$t = 3 \quad t = 5$

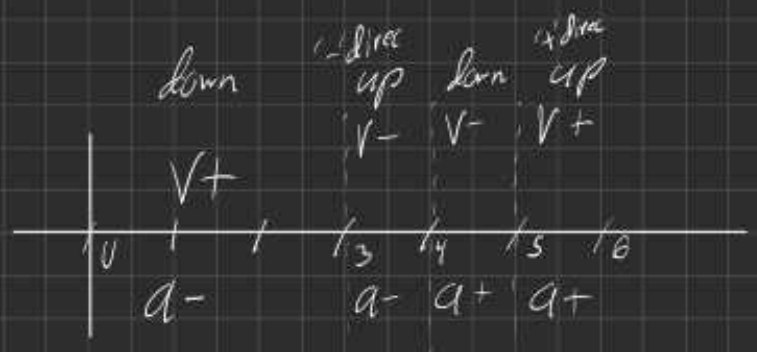
Where is it speeding up & slowing down?

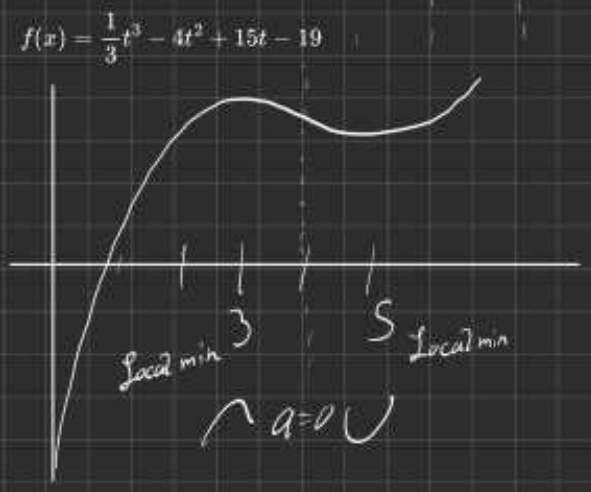
$V+ \; V-$ speeding up
$a+ \; a-$

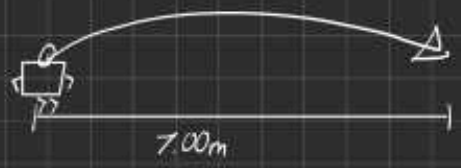
$V+$ or $V-$ slowing down
$a- \; a+$

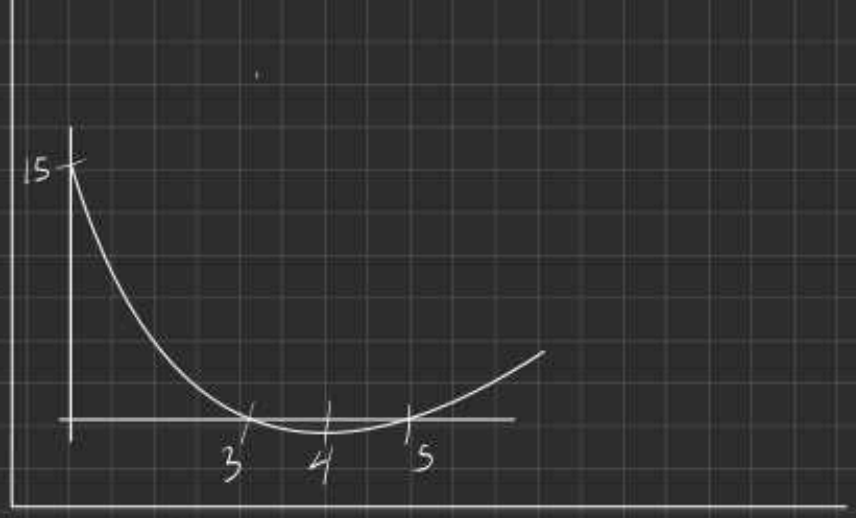
$V(t) = t^2 - 8t + 15$
$t = 5 \quad t = 3$

$a(t) = 2t - 8$
$2(t-4)$









symmetrical motion

max motion for symm motion is 45°

If she jumps at 45° what is V_I

range x uniform

$\Delta x = V_0 \cos\theta t$

$V_x = V_0 \cos\theta$

UAM y

$\Delta y = -\frac{1}{2} g t^2 + V_0 \sin\theta t$

$V_y = -gt + V_0 \sin\theta$

$V_y^2 = (V_0 \sin\theta)^2 - 2g\Delta y$

$\Delta y = -\frac{1}{2} g t^2 + V_0 \sin\theta t = 0$

$\frac{1}{2} g t^2 = V_0 \sin\theta t$

$t = \frac{2 V_0 \sin\theta}{g}$ Only for symm. motion

Range

$\Delta y = V_0 \cos\theta t = 7m$

$V_0 \cos\theta \cdot \frac{2 V_0 \sin\theta}{g} = 7m$

$V_0^2 \left(\frac{2 \sin\theta \cos\theta}{g}\right) = 7m$

$\frac{V_0^2 (\sin 2\theta)}{g} = 7m$

$V_0 = \sqrt{7 \cdot 9.81}$

$V_0 = 8.29 m/s^2 \cdot 2.24 = 18.56 mph$

What is max height

$\Delta y_{max} = -\frac{1}{2} g t_{1/2}^2 + V_0 \sin\theta t_{1/2}$

$= -\frac{1}{2} g \left(\frac{2 V_0 \sin\theta}{g} \cdot \frac{1}{2}\right)^2 + V_0 \sin(45)\left(\frac{2 V_0 \sin\theta}{g} \cdot \frac{1}{2}\right)$

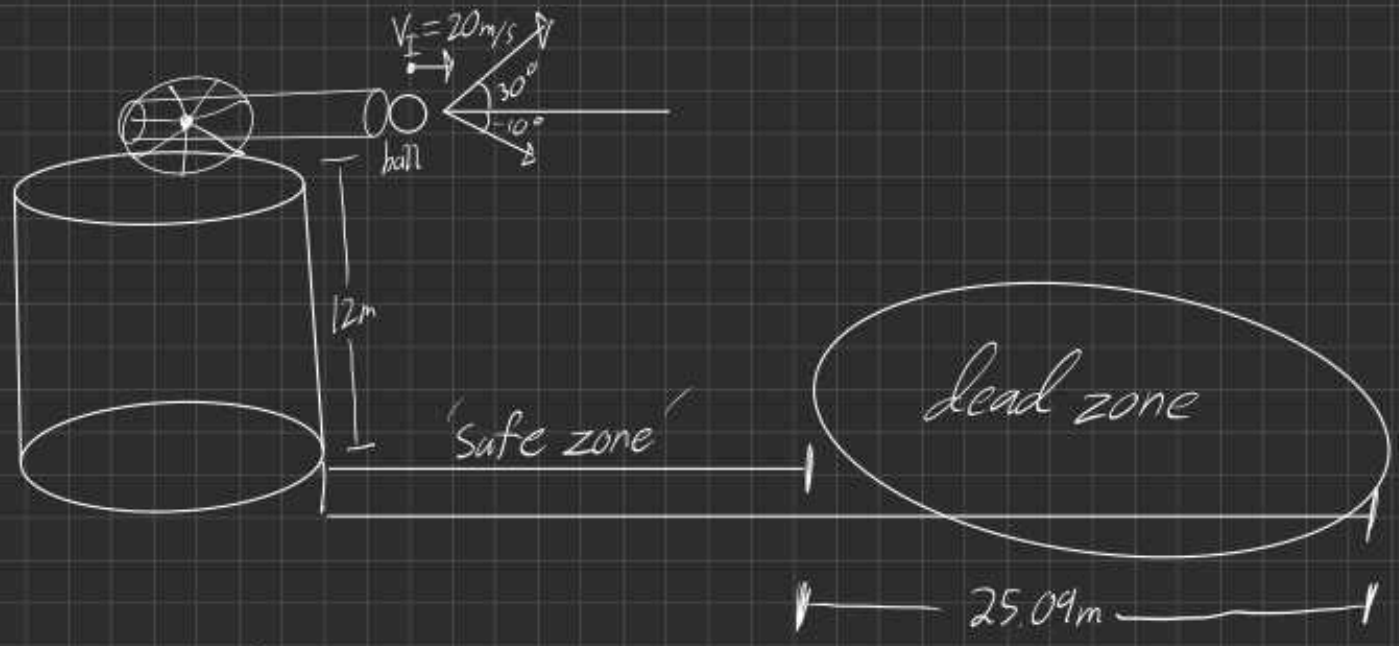
$= 4.905 \left(\frac{8.29 \cdot \sin(45)}{9.81}\right)^2 + 8.29 \sin(45)\left(\frac{8.29 \cdot \sin(45)}{9.81}\right)$

$= 4.905 \cdot 0.3571 + 5.8619 (0.5475)$

$= 1.7513 + 3.5028$

$= 5.24m$

$= 1.75m$



$\{-10^\circ\}$

$$\Delta y = \frac{1}{2}(-g)t^2 + V_0 \sin(\theta)t$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$0 = \frac{1}{2}(-g)t^2 + V_i \sin(\theta)t - \Delta y$$

$$0 = -4.905t^2 + \underset{-3.47}{20\sin(-10)}t - 12$$

$$t = \frac{+3.47 \pm \sqrt{(-3.47)^2 + 4(+4.905)(+12)}}{2(-4.905)}$$

$$t = \frac{+3.47 \pm \sqrt{12.0409 + 235.44}}{-9.81}$$

$$t = \frac{+3.47 \pm \sqrt{247.48}}{-9.81}$$

$$t = \frac{3.47 - 15.73}{-9.81} \qquad t = \frac{3.47 + 15.73}{-9.81}$$

$t = 1.25\,s$ $\qquad$ $t =$ imaginary

$$t = \frac{-10 \pm \sqrt{(10)^2 + 4(+4.905)(+12)}}{2(-4.905)}$$

$$t = \frac{-10 \pm 18.32}{-9.81}$$

$+$ $\qquad$ $-$

$t =$ imaginary $\qquad$ $t = 2.87\,s$

$$\Delta x = V_0 \cos\theta t$$

$$\Delta x = 20\cos(-10)\cdot(1.25)$$

$$\Delta x = 24.62\ m$$

$$\Delta x = V_0 \cos\theta t$$
$$= 20\cos(30)(2.87)$$

$$\Delta x = 49.71\ m$$

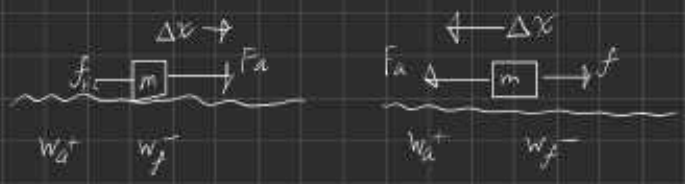
Conservative vs. Non conservative Forces

1. can derive a PE Function
2. Path independent
3. Net Work for round trip path = 0
4. No loss in mechanical energy
   KE + PE

$W_g = \int F_g \, dy$
$= mg \int dy$
$= mg(y_f - y_i)$
$= mg\Delta y$

$\Delta PE_g = mgy_f - mgy_i$

Gravity – Conservative



$W_{net} = W_{cons} + W_{non\ cons}$

$\Delta KE = -\Delta PE +$

$\Delta KE + \Delta PE = W_{non\ cons} = \Delta E$

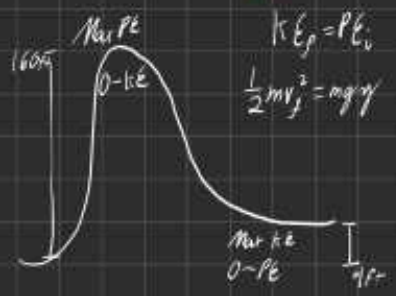
change in mechanical energy

Steel Eel ignore friction ∴ $W_{nc} \approx 0$

$\Delta E = \Delta KE + \Delta PE = W_{nc} \approx 0$

$KE_f - KE_i + PE_f - PE_i \approx 0$

$\Delta y = 160 - 4$
$\Delta y = 156 ft$
$= 47.55 m$

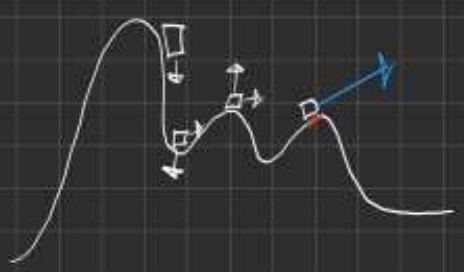


$KE_f = PE_i$

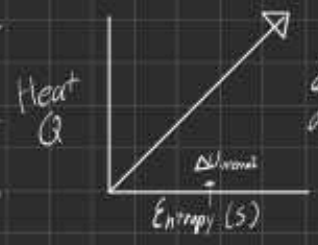
$\frac{1}{2}mv_f^2 = mgy$

$V_f = \sqrt{2g \cdot 47.55m}$

$V_f \approx 30.54 m/s \cdot \frac{2.24 mph}{1\ m/s} \approx 70 mph$



Heat → measurable → Thermometer °K °F °C



$\frac{dQ}{dS} = \text{Temperature} = T$

Pressure

$P = \frac{F}{A} \frac{dl}{dl}$

$P = \frac{dW}{dV}$

phonons → vibrations in a 3D lattice copper

$dQ = T dS$

$dW = P dV$

2 ways to change internal energy.

1) Add/Remove $Q \rightarrow dQ$
2) Do work $\rightarrow dW$

$\Delta U_{int} = dQ - dW$

$\Delta U_{int} = TdS - PdV$

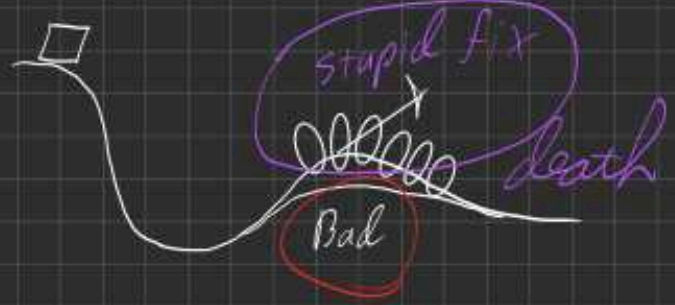
1st Law of Thermodynamics

In a closed System: No Q in/out of system

$\Delta U_{int} = dQ - dW$

$\Delta U_{int} = -W_{nc}$

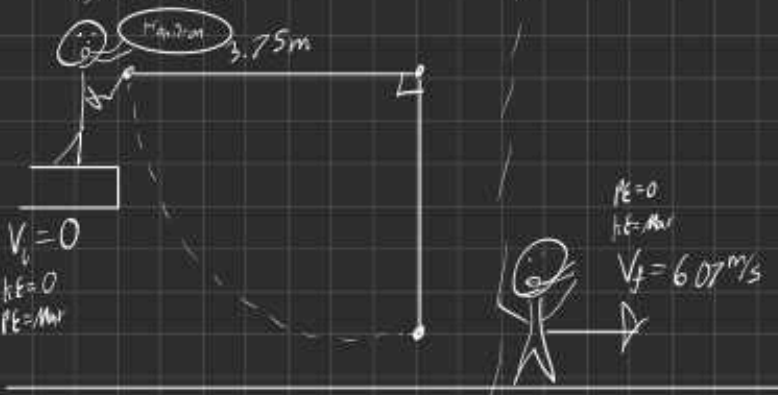
If $W_{nc} \approx 0$ then $\Delta U_{int} \approx 0$



In a Broadway performance, an 80kg actor steps off a ledge holding a 3.75m cable and swings down to the floor. Then, he lets go of the rope and slides to a stop.

Calculate speed when he hits the floor?

How much stage length is needed?
w/ $\mu = 0.3$ fric



τ_{fg} not constant $= r \times F$
$= I\alpha$

$KE = \frac{1}{2}mv^2$

$PE = mgh$
or
Δy

Cons. Energy

$W_{nc} = -\Delta U_{int}$

$\Delta E_{tot} = 0$

$0 = \Delta KE + \Delta PE + \Delta U_{int}$

From all sources

$\Delta KE_T + \Delta KE_R + \Delta PE_g = 0$

$KE_{fT} - KE_{iT} + KE_{fR} - KE_{iR} + PE_f - PE_i = 0$

$KE_{fT} + KE_{fR} - PE_i = 0$

$KE_{fT} + KE_{fR} = PE_i$

$\frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2 = mgh$

$v = r\omega$
$\omega = \frac{v}{r}$

$\frac{1}{2}mv_f^2 + \frac{1}{2}mr^2\left(\frac{v}{r}\right)_f^2 = mgh$

$\frac{1}{2}v^2 + \frac{1}{2}v^2 = gh$

$v_f^2 = gh$

$v_f = \sqrt{gh}$

$V_f = \sqrt{9.81 m/s^2 \cdot 3.75m}$

$V_f = \sqrt{36.79}$

$V_f = 6.07 m/s \cdot 2.24 \frac{mph}{m/s} = 13 mph$

Similar on Exam

Le Grangean ?
Classical mechanics

Δx	✗
V_0	✓
V_f	✓
t	✗
a	✗

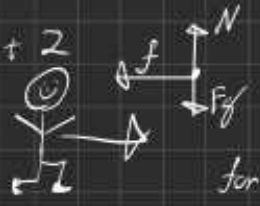
Use work energy Theorem!!!

$\int F\,dx = W = \Delta KE$

work

This force is friction to change his KE

Part 2



$N = F_g$

$-F_f \cdot \Delta x = \Delta KE$

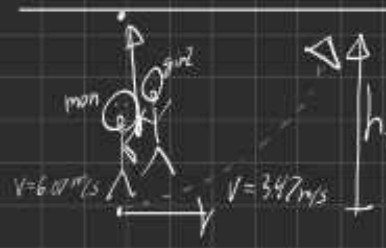
$\mu \cdot -N \cdot \Delta x = KE_f - KE_i$

$\mu(-mg)\Delta x = -\frac{1}{2}mV_i^2$

$\Delta x = \frac{\frac{1}{2}V_i^2}{\mu g} = \frac{\frac{1}{2}\cdot(6.07)^2}{0.3 \cdot 9.82} = 6.26m$

for horizontal $F_g = N$

stage tot $= 3.75m + 6.26m \approx 10m \approx 30ft$



$\Delta p = 0$

$p_f - p_i = 0$

$p_f = p_i$

$(\vec{p}_1 + \vec{p}_2)_f = (\vec{p}_1 + \vec{p}_2)_i$

$m_1V_1 + m_2V_2 = (m_1V_1)_i$

$V_f(m_1 + m_2) = (m_1V_1)_i$

$V_f = \frac{(m_1V_1)_i}{m_1 + m_2}$

$V_f = \frac{(80 \cdot 6.07)}{140 kg} = 3.47 m/s$

$p_1 = man \rightarrow 80 kg$

$p_2 = girl \rightarrow 60 kg$

$V_{1f} = V_{2f}$

$V_i^2 = gh$

$h = \frac{V_i^2}{g}$

$h = \frac{12}{9.81}$

$h = 1.23m$

$KE_{fT} - KE_{iT} + KE_{fR} - KE_{iR} + PE_f - PE_i = 0$

$PE_f = KE_{iT} + KE_{iR}$

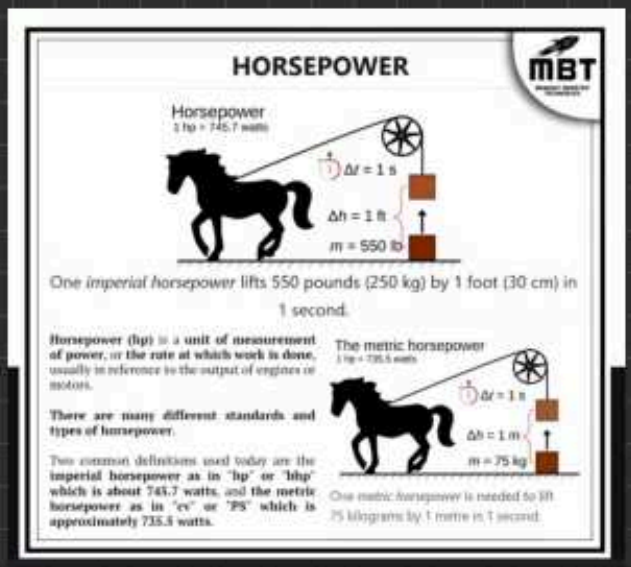
Power

$P = \frac{W}{\Delta t}$ or $F_x v_x$

$1hp = \frac{33{,}000 lbs}{min} = \frac{550 lbs}{sec} = 745.7 Nm/s$

Imperial hp = 745.7 watts SI

~~Metric hp = 735.5 watts~~



Angular momentum

$\Delta \vec{L} = 0$

$L_f - L_i = 0$

$I_f \omega_f = I_i \omega_i$

Newt's 2nd Law

Linear $F = ma = m\frac{dv}{dt} = \frac{d\vec{p}}{dt}$

Rotational $\tau = I\alpha = m\frac{d\omega}{dt} = \frac{d\vec{L}}{dt}$

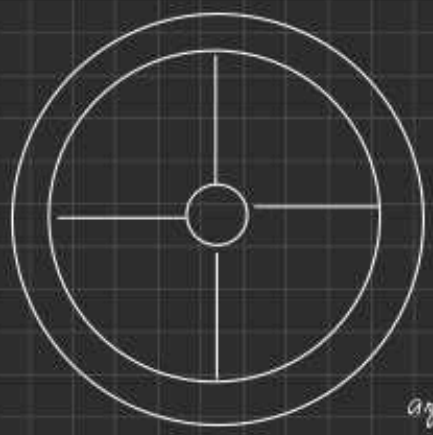
$\vec{p} = m\vec{v}$

$\vec{L} = I\vec{\omega}$

$\omega = \frac{d\theta}{dt} = \frac{1 \text{ rotation}}{\Delta t_1} = \frac{2\pi \text{ rad}}{\Delta t_1}$

$\frac{2 \text{ rot}}{\Delta t_2} = \frac{2 \cdot 2\pi \text{ rad}}{\Delta t_2}$

$\omega = \frac{2\pi \text{ rad} \cdot \#\text{ of turns}}{\Delta t}$



$F_c = F_g = 1G = m \cdot 9.81 m/s^2$, $\omega = 2\pi \cdot f$

frequency (f)

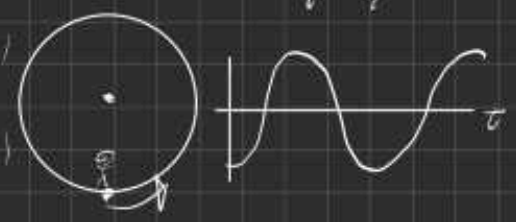
$\frac{mv^2}{r} = m9.81$

$v = \sqrt{r \cdot 9.81}$

$v = \sqrt{950m \cdot 9.81 m/s^2}$

$v = 96.5 m/s$

angular speed $\omega = \frac{v}{r} = 0.1016 \frac{rad}{s}$ (constant)



Period $(T) = \frac{1}{f} = \frac{\Delta t}{\#\text{ of turns}}$

$v = r\omega = r2\pi f = \underbrace{2\pi r}_{\text{Circumference}} \cdot f = Cf$

If 100 people move from rim to center.

Is there a significant change in ω?

$I_i \omega_i = I_f \omega_f$

$\omega = \boxed{\frac{I_i}{I_f}} \omega_i$

how significant is this

$I_i = I_{population} + I_{station}$

$I_f = I_{new\ pop} + I_{station}$

$I = 5 \times 10^8 kgm^2$

$pop = 10^6$

No real change

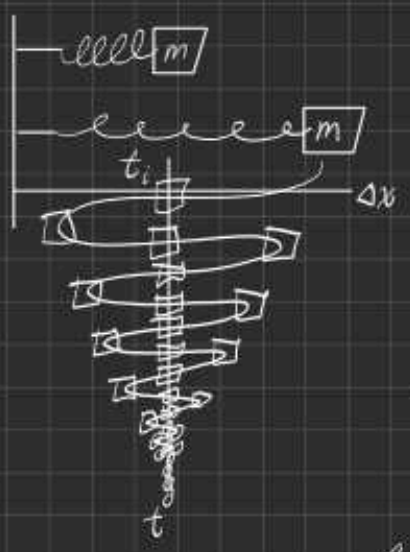
$s = r\theta$

$\Delta s = r\Delta\theta$

Springs & Hooke Law

$$F_s = -k\Delta x = ma_s$$

$$W_s = \int F\,dx$$
$$= \int_{x_i}^{x_f} -kx\,dx$$
$$= -k\int_{x_i}^{x_f} x\,dx$$
$$= -\frac{k}{2}x^2\Big|_{x_i}^{x_f}$$
$$= -\left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right)$$
$$W_s = PE_{s_f} - PE_{s_i}$$
$$= -\Delta PE$$



PE_{max} @ full stretch / compression
$KE = 0$

KE_{max} @ equilibrium

making the func.

$$\underbrace{F_s = -k\Delta x = ma_s}$$

$x(t), v(t), \& a(t)$?

$$a = -\frac{k}{m}\Delta x \qquad \text{Let } \frac{k}{m} = \omega^2$$

$$\frac{dv}{dt} = \boxed{\frac{d^2x}{dt} = -\omega^2 x}$$

$v = r\omega$

$\omega = \frac{d\theta}{dt}$

$d\theta = \omega\,dt$

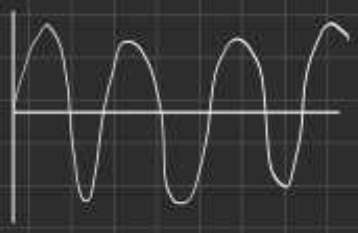
consider a function that satisfies the relationship

Phase angle/initial $\angle = 0$ @ x_{max} & $t=0$ then

$$x(t) = A\cos(\omega t + \theta)$$

Amplitude
x_{max}

$$V(t) = -A\sin(\omega t + \theta)\cdot\omega$$
$$a(t) = -A\cos(\omega t + \theta)\cdot\omega^2$$



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initial: $\omega t + \theta$

final: $\omega(t+T) + \theta$

$$= \omega(t+T) + \theta - (\omega t + \theta)$$
$$= \omega t + \omega T + \theta - \omega t - \theta$$
$$\Delta\theta = 2\pi = \omega T \Rightarrow T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$\omega = 2\pi f \qquad \omega^2 = k/m$

$$\sqrt{k/m} = 2\pi f$$
$$f = 2\pi\sqrt{m/k}$$

$$\Delta KE = -\Delta PE$$

$$E_{tot} = \underbrace{KE + PE}_{\text{interplay while oscillation occurs}}$$

tot PE during

$$\frac{1}{2}kx_{max}^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$v^2 = \frac{kA^2 - kx^2}{m}$$

$$v(x) = \sqrt{\frac{k}{m}(A^2 - x^2)} \rightarrow v(x) = \pm\omega\sqrt{A^2 - x^2}$$

Period

$$T = \frac{2\pi}{\omega}$$

Circular motion

$v = r\omega$ $\quad \omega = \frac{d\theta}{dt}$

Angular speed

$\omega = \theta / t$

Conversion

Linear → Angular

$\frac{\Delta v}{\Delta t} = r\frac{\Delta \omega}{\Delta t}$

$a_{tang.} = r\alpha$

$a_{net} = \sqrt{a_c^2 + a_{tang}^2}$

$1\ rev = 2\pi\ rad = 360°$

SI

total arc length → $\Delta S = r\Delta\theta$ → angular distance

(m) (m) (rad)

radius or rotation

$\Delta\theta = \frac{\Delta s\ m}{r\ m}$

radians is dimensionless

$a_c = \frac{v^2}{r}$ $\quad a_c = r\omega^2$ $\quad a_t = r\alpha$

Angular position ≡ rad

ω Velocity $= \frac{d\theta}{dt} = \frac{rad}{s} = \frac{1}{s} = s^{-1}$ or rad/s

α acceleration $= \frac{d\omega}{dt} = \frac{rad}{s^2} = \frac{1}{s^2} = s^{-2}$ or rad/s^2

Linear	Rotational kinem.
$s = r\theta$	$\omega_f = \omega_i + \alpha t$
$v_t = r\omega$	$\theta = \frac{1}{2}(\omega_i + \omega_f)t$
$a_t = r\alpha$	$\theta = \omega_i t + \frac{1}{2}\alpha t^2$
	$\omega_f^2 = \omega_i^2 + 2\alpha\theta$

Linear vs. Angular quantities

Quantity	Linear	Conversion	Angular
Position	x and/or y	$x = r\cos\theta$ $y = r\sin\theta$ $s = r\theta$	θ
Displacement	$\Delta x = x - x_0$ $\Delta y = y - y_0$	$\Delta s = r\,\Delta\theta$	$\Delta\theta = \theta - \theta_0$
Velocity	$v_x = \Delta x/\Delta t$ $v_y = \Delta y/\Delta t$	$v_{tangential} = r\,\omega$	$\omega = \Delta\theta/\Delta t$
Acceleration	$a = \Delta v/\Delta t$	$a_{tangential} = r\,\alpha$	$\alpha = \Delta\omega/\Delta t$

Linear vs Angular kinematics

Equation	Linear	Conversion	Angular
Position	x and/or y	$x = r\cos\theta$ $y = r\sin\theta$	θ
Displacement	$\Delta x = \frac{1}{2}at^2 + v_0 t$		$\Delta\theta = \frac{1}{2}\alpha t^2 + \omega_0 t$
Velocity	$v = at + v_0$	$v_{tangential} = r\,\omega$	$\omega = at + \omega_0$
Acceleration	$a = a_0$	$a_{tangential} = r\,\alpha$	$\alpha = \alpha_0$
Time Indep.	$v^2 = v_0^2 + 2a\Delta x$		$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$

$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$W = \Delta KE$

$W_{gravity} = -\Delta PE_g$

$PE_{spring} = \frac{1}{2}kx^2$

$W_s = -\left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right) = -\Delta PE_s$

$W_{conservative} = \Delta KE = -\Delta PE$

$W_r = I\Delta\theta$

$W_r = \Delta KE_r$

$W_{non\text{-}conser.} = -\Delta U$

$W_{non\text{-}cons.} = -\Delta U = \Delta E_{tot}$

Work Energy Theorem

$W_{nc} = -\Delta U = \Delta KE$

Angular Momentum ($\mathcal{L}$)

$\mathcal{L}_1 = \mathcal{L}_2$

$I_1\omega_1 = I_2\omega_2$

Linear Momentum

$J = \Delta p$

Impulse (I)

$I = F\Delta t = m\Delta v$

Momentum (P)

$\vec{p} = m\vec{v}$

$p_i = p_f$

$(p_1 + p_2)_i = (p_1 + p_2)_f$

$m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$

$x: m_1v_{1i} + 0 = m_1v_{1f}\cos\theta + m_2v_{2f}\cos\phi$

$y: m_1v_{1i} + 0 = m_1v_{1f}\sin\theta + m_2v_{2f}\sin\phi$

Power

$P = \frac{W}{t}$

Potential Energy

$PE = mgh$

Kinetic Energy

$KE = \frac{1}{2}mv^2$

$W_{net} = \Delta KE = KE_f - KE_i$